

Exact POMDP Solutions: α -vectors

Recap

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- POMDP

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$$(S, A, O, R, T, Z, \gamma)$$

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

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$$b_t(s) = P(s_t = s \mid h_t)$$

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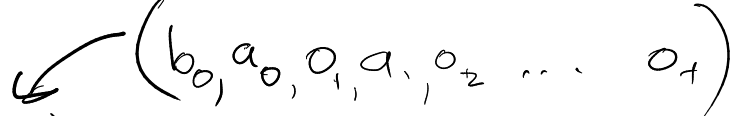
$$b_t(s) = P(s_t = s \mid h_t)$$

$$b' = \tau(b, a, o)$$

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- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

$$b_t(s) = P(s_t = s \mid h_t)$$

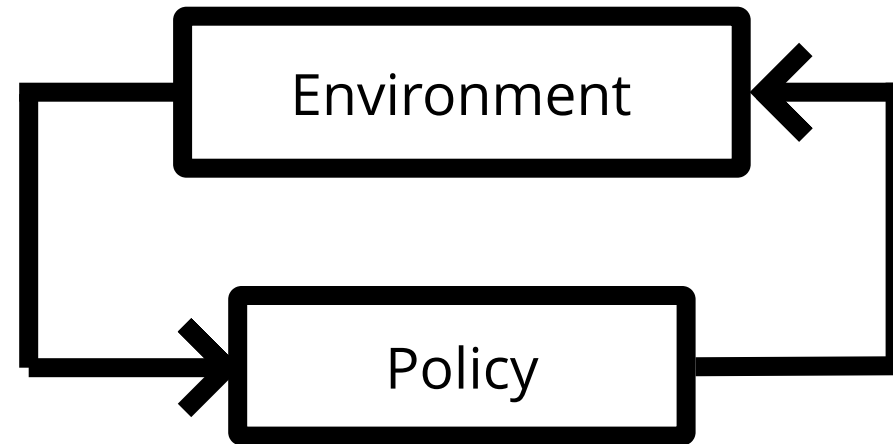
 $(b_0, a_0, o_1, a_1, o_2, \dots, o_t)$

The handwritten tuple $(b_0, a_0, o_1, a_1, o_2, \dots, o_t)$ is written above the equation. An arrow points from the h_t in the equation to the b_0 in the tuple.

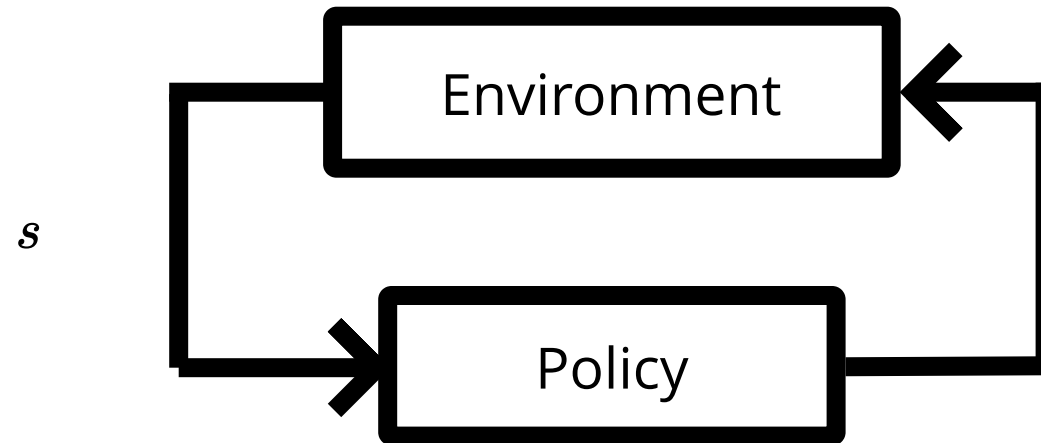
$$b' = \tau(b, a, o)$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

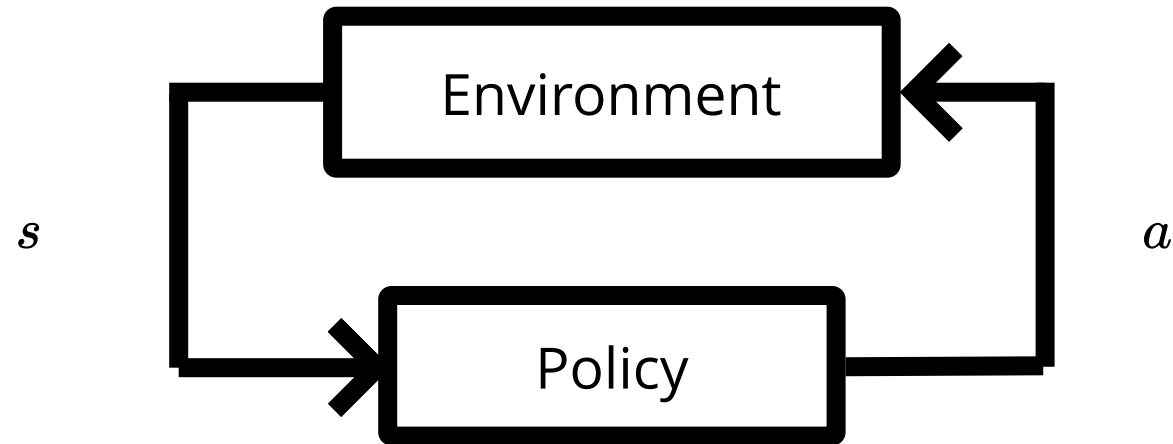
MDP Sense-Plan-Act Loop



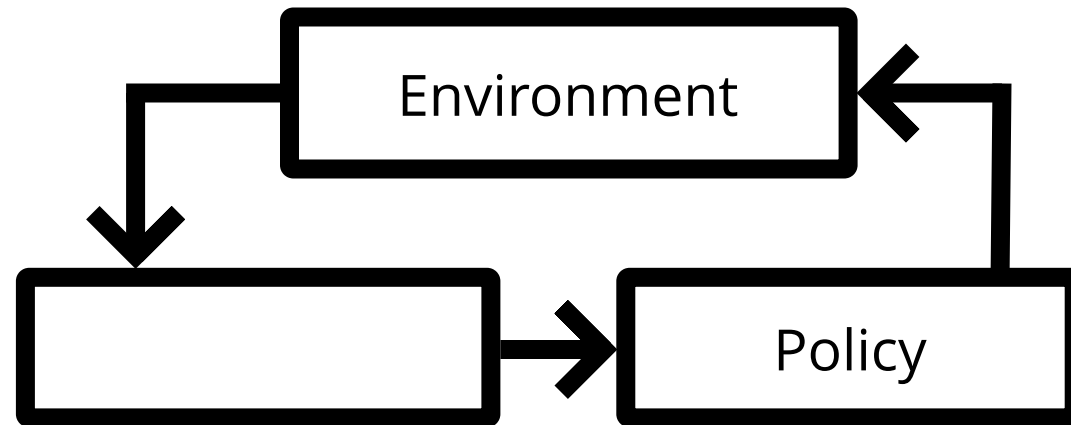
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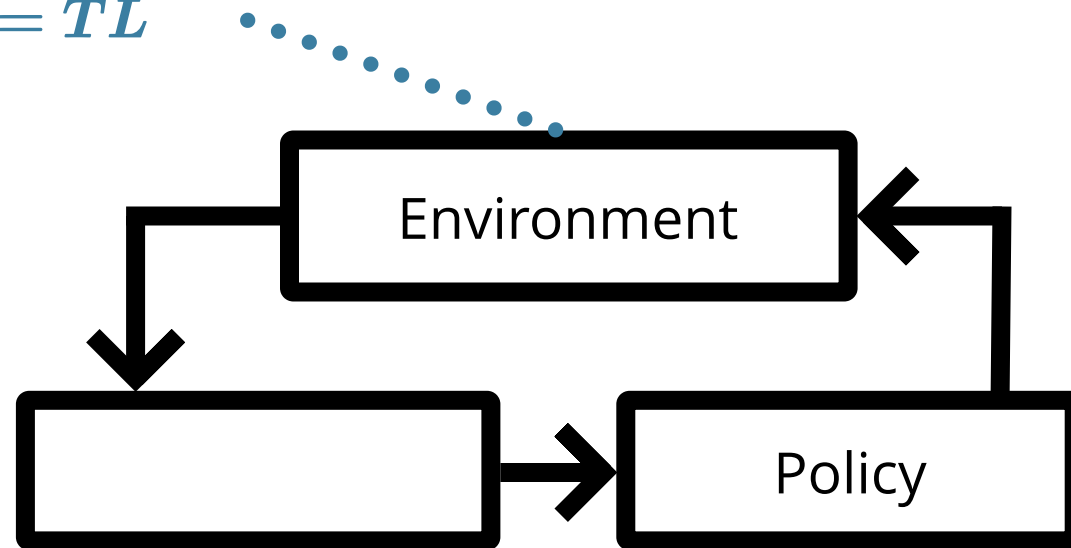
POMDP Sense-Plan-Act Loop



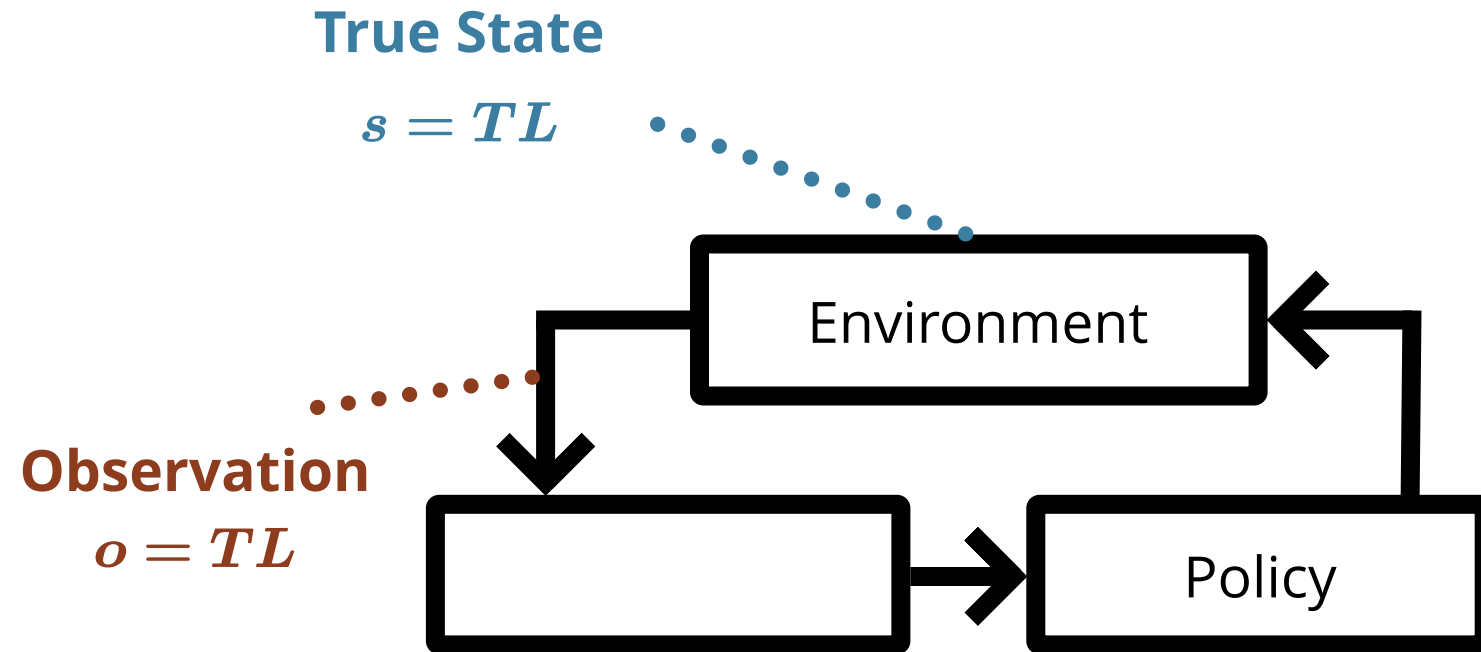
POMDP Sense-Plan-Act Loop

True State

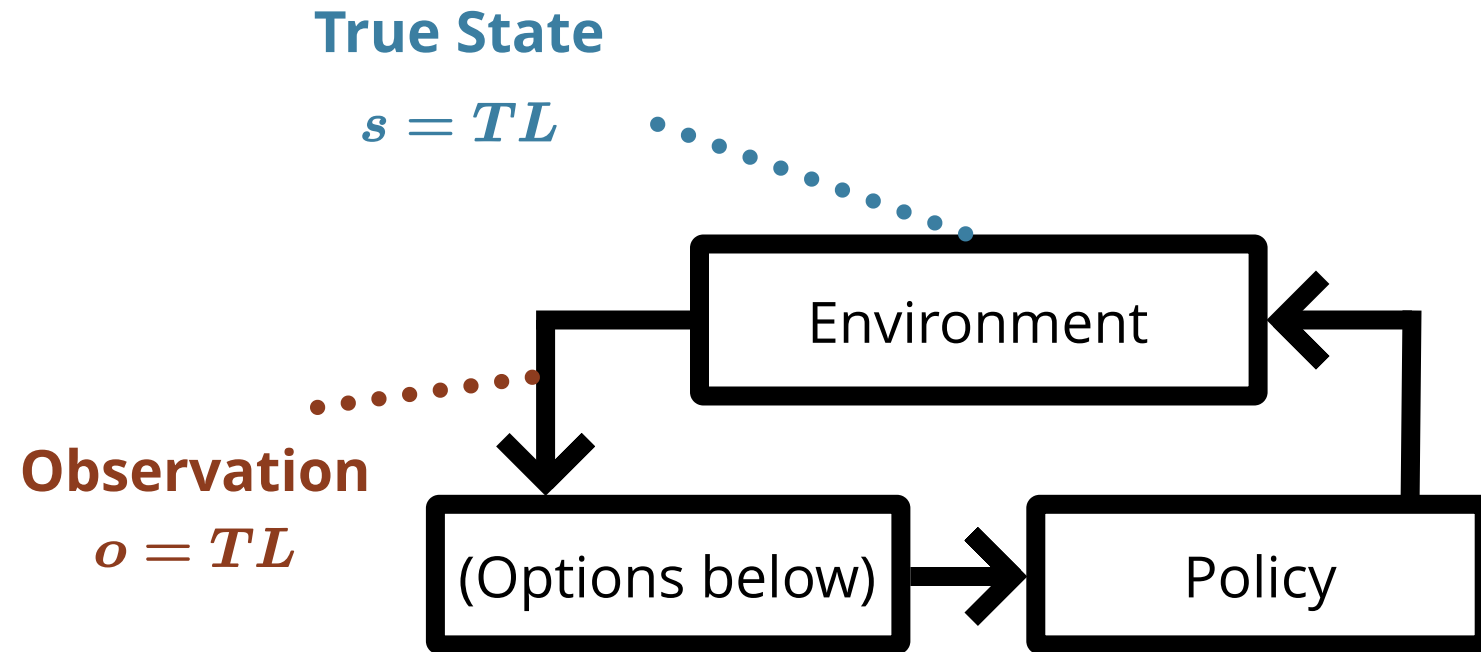
$$s = TL$$



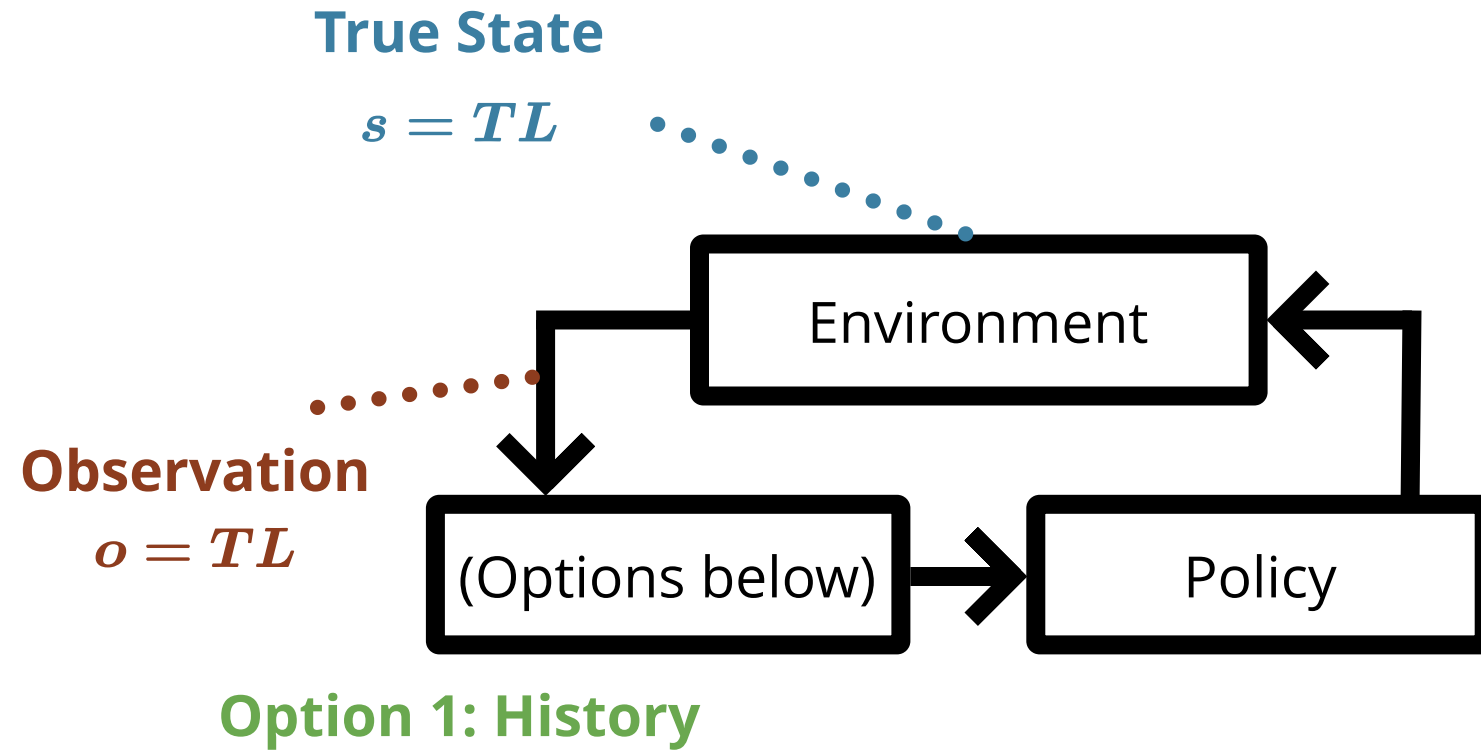
POMDP Sense-Plan-Act Loop



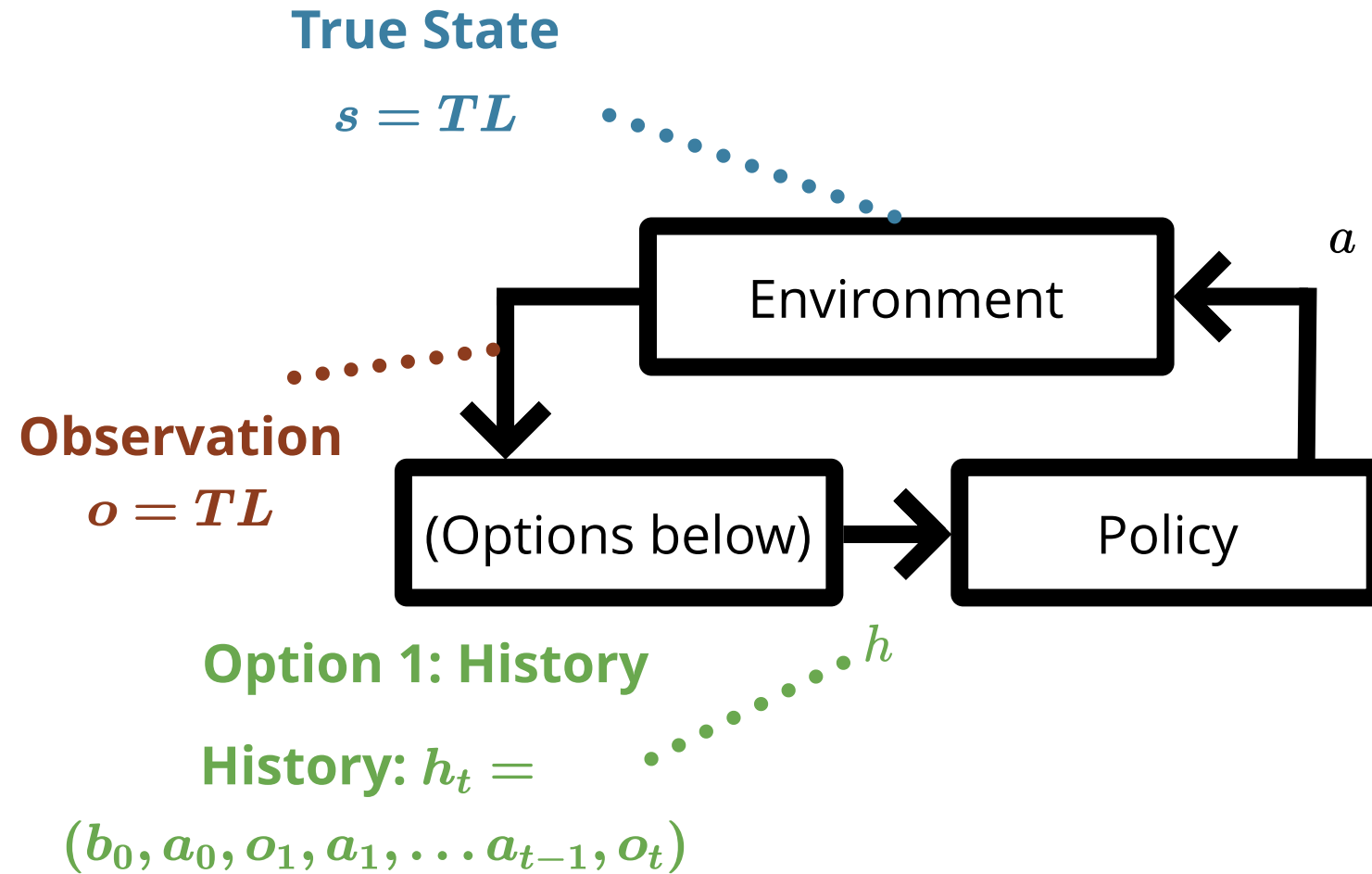
POMDP Sense-Plan-Act Loop



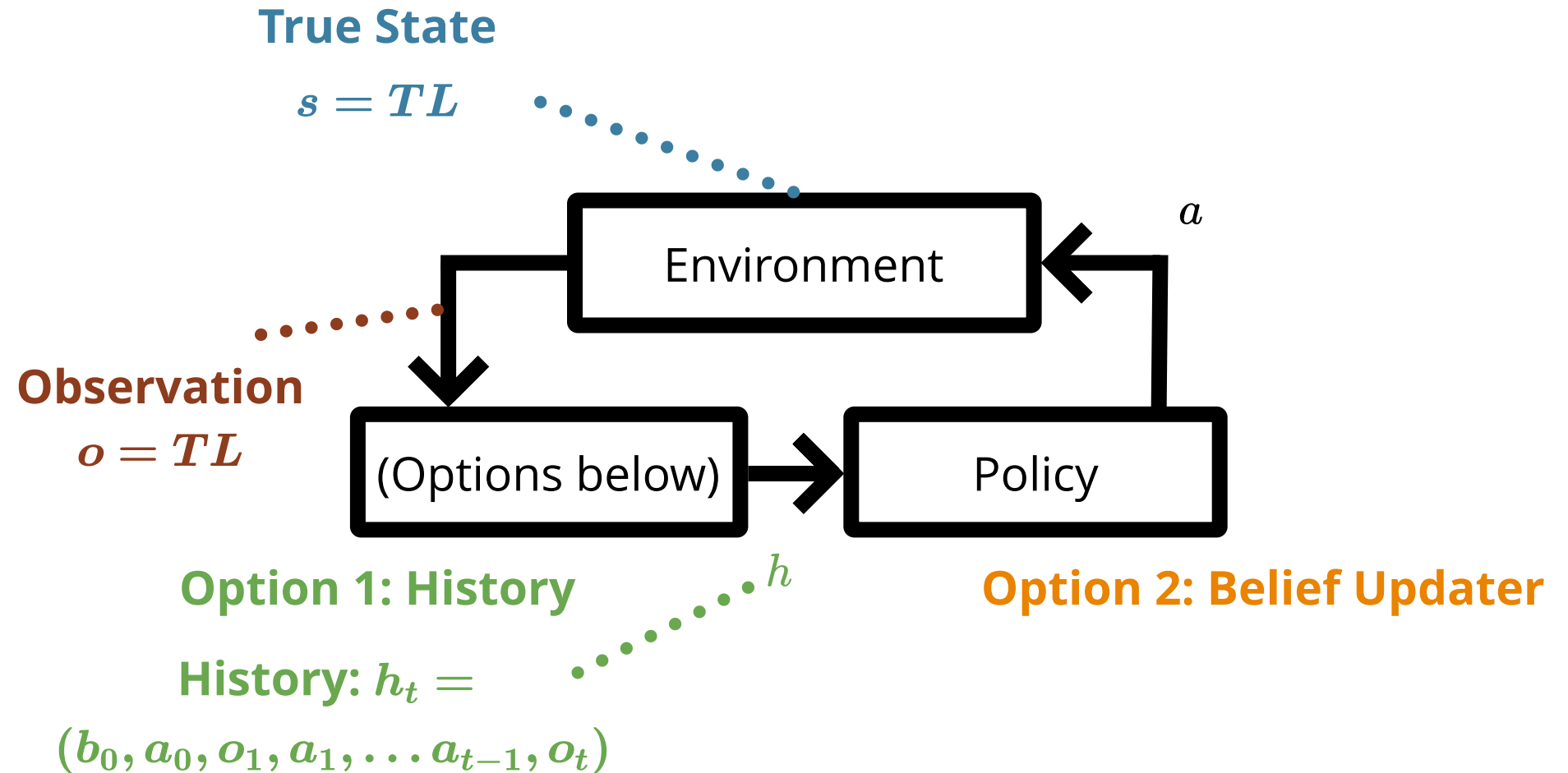
POMDP Sense-Plan-Act Loop



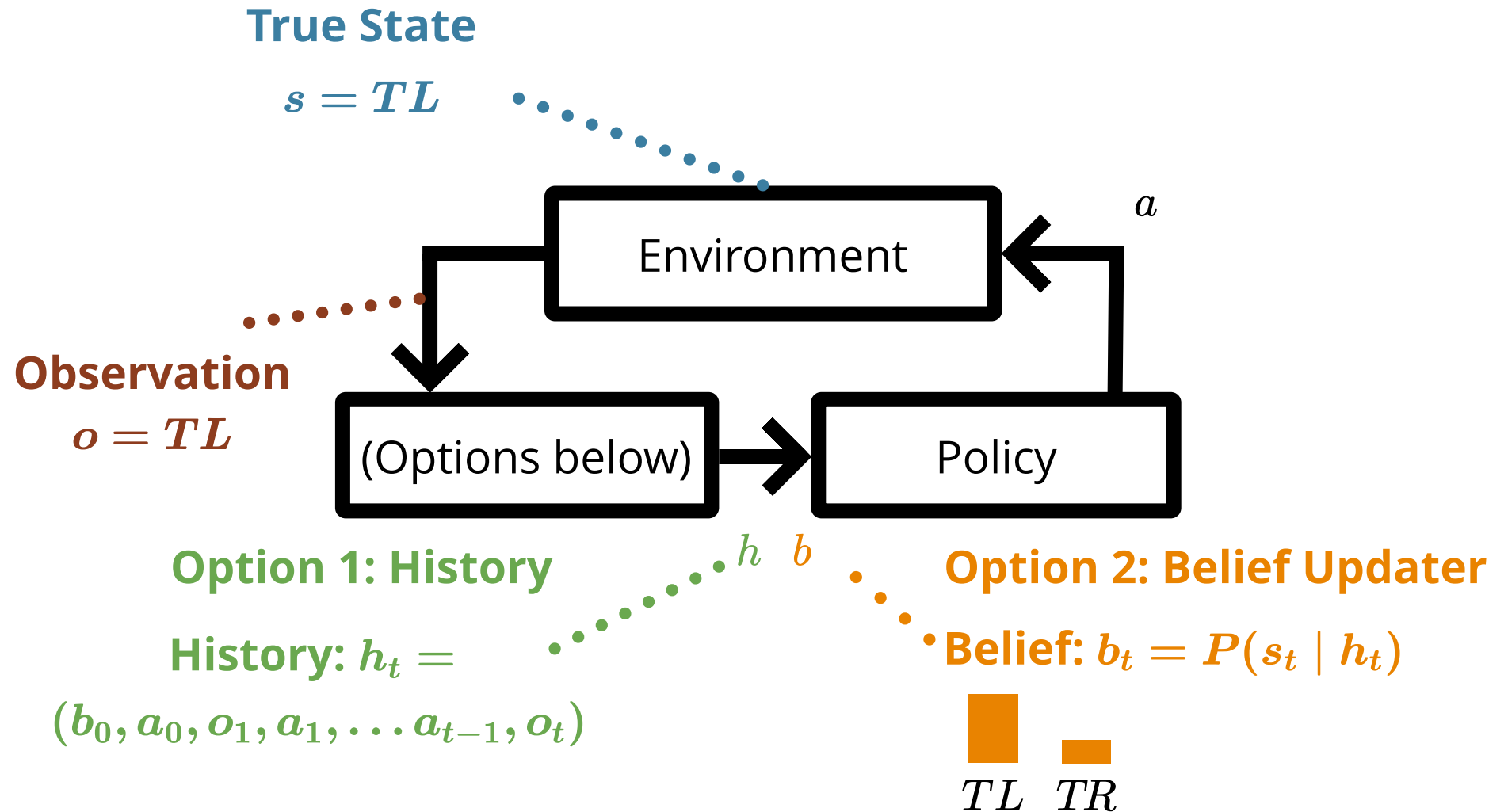
POMDP Sense-Plan-Act Loop



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POMDP Sense-Plan-Act Loop



Exercise 1: Crying Baby Belief Update

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$$S = \{h, \overset{\text{not}}{\neg}h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

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Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

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$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

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$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

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$$T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\}$$

$$T(\neg h \mid \cdot, f) = 1.0$$

$$b'(s') \propto \underline{Z(o \mid a, s')} \sum_s T(s' \mid s, a) b(s)$$

Starting at a $b(h) = 0$, calculate

b' with $a = \neg f$ and $o = c$.

$$R(s, a) = R(s) + R(a)$$

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$$Z(c \mid \cdot, h) = 0.8$$

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$$\gamma = 0.9$$

$$b'(h) \propto Z(c \mid \neg f, h) (T(h \mid h, \neg f) b(h) + T(h \mid \neg h, \neg f) b(\neg h))$$

$$0.8 \quad (1.0 \quad 0.0 + 0.1 \quad 1.0)$$

$$\rightarrow b'(h) \propto 0.08$$

$$b'(\neg h) \propto Z(c \mid \neg f, \neg h) (T(\neg h \mid h, \neg f) b(h) + T(\neg h \mid \neg h, \neg f) b(\neg h))$$

$$0.1 \quad (0 \quad 0.9 \quad 1.0)$$

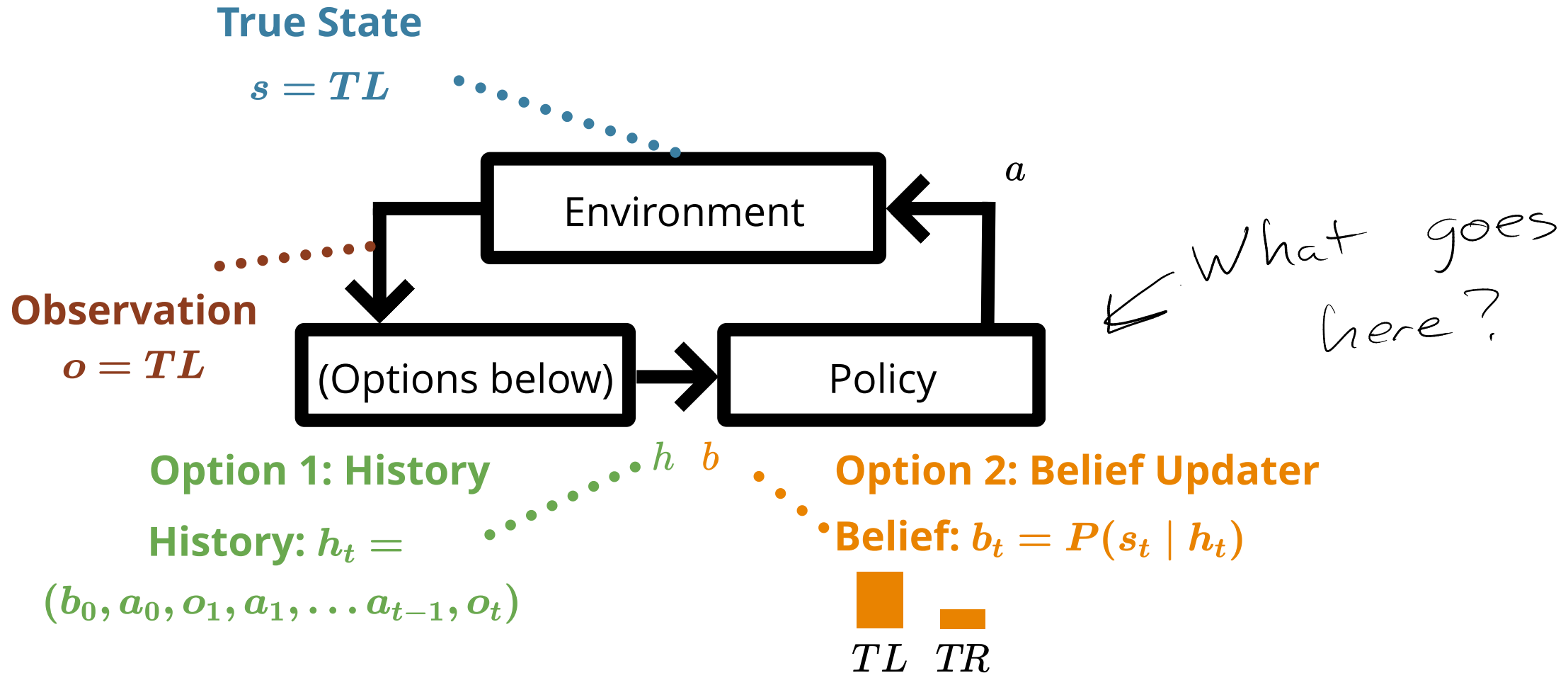
$$\rightarrow b'(\neg h) \propto 0.09$$

$$b'(h) = \frac{0.08}{0.08 + 0.09} = 47\%$$

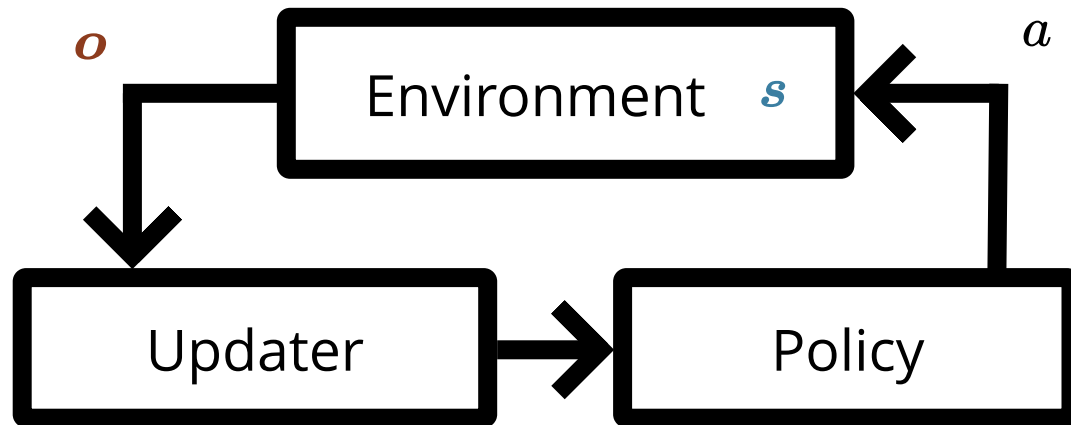
$$b'(\neg h) = 1 - 47\% = 53\% = \frac{0.09}{0.08 + 0.09}$$

Guiding Question

How do we calculate the optimal action in a POMDP?



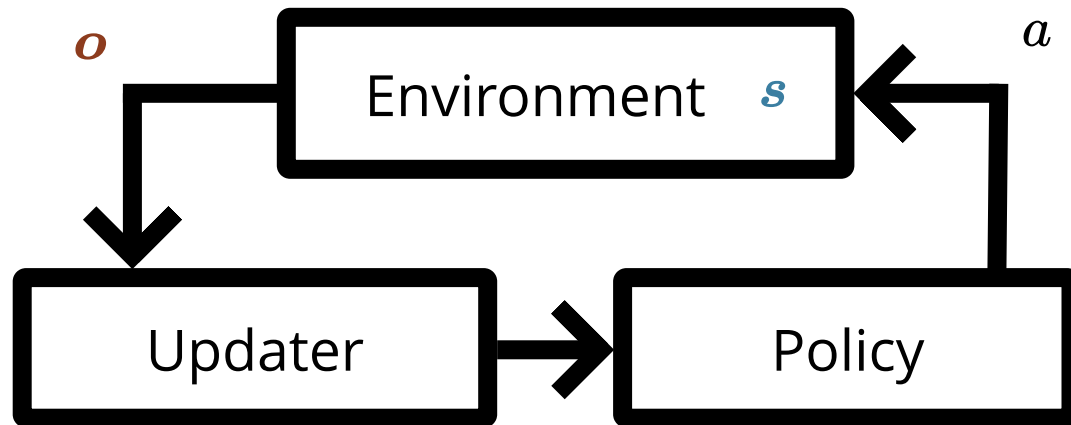
Belief-Space MDP



$$b_t = P(s_t | h_t)$$

 TL  TR

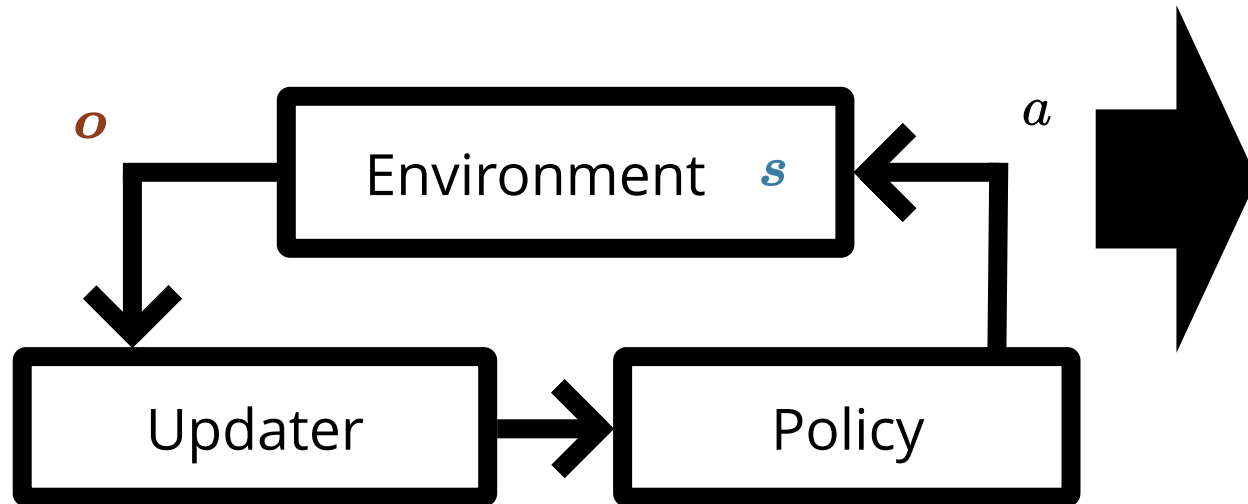
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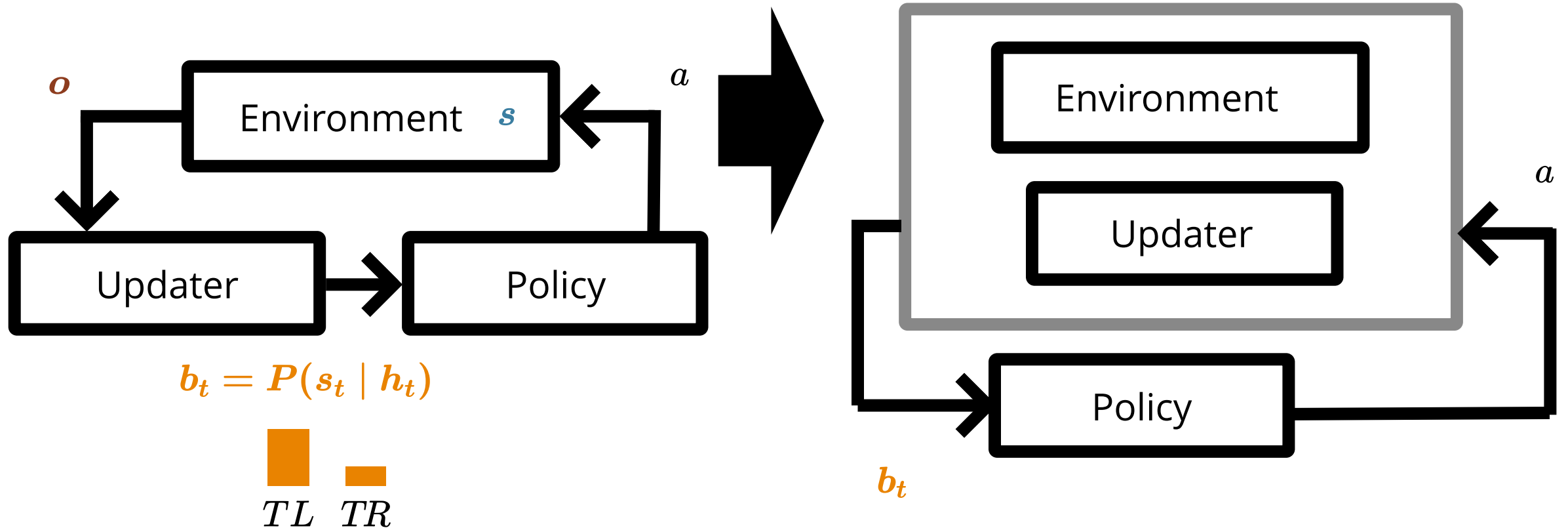
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Belief Dynamics

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Belief Dynamics

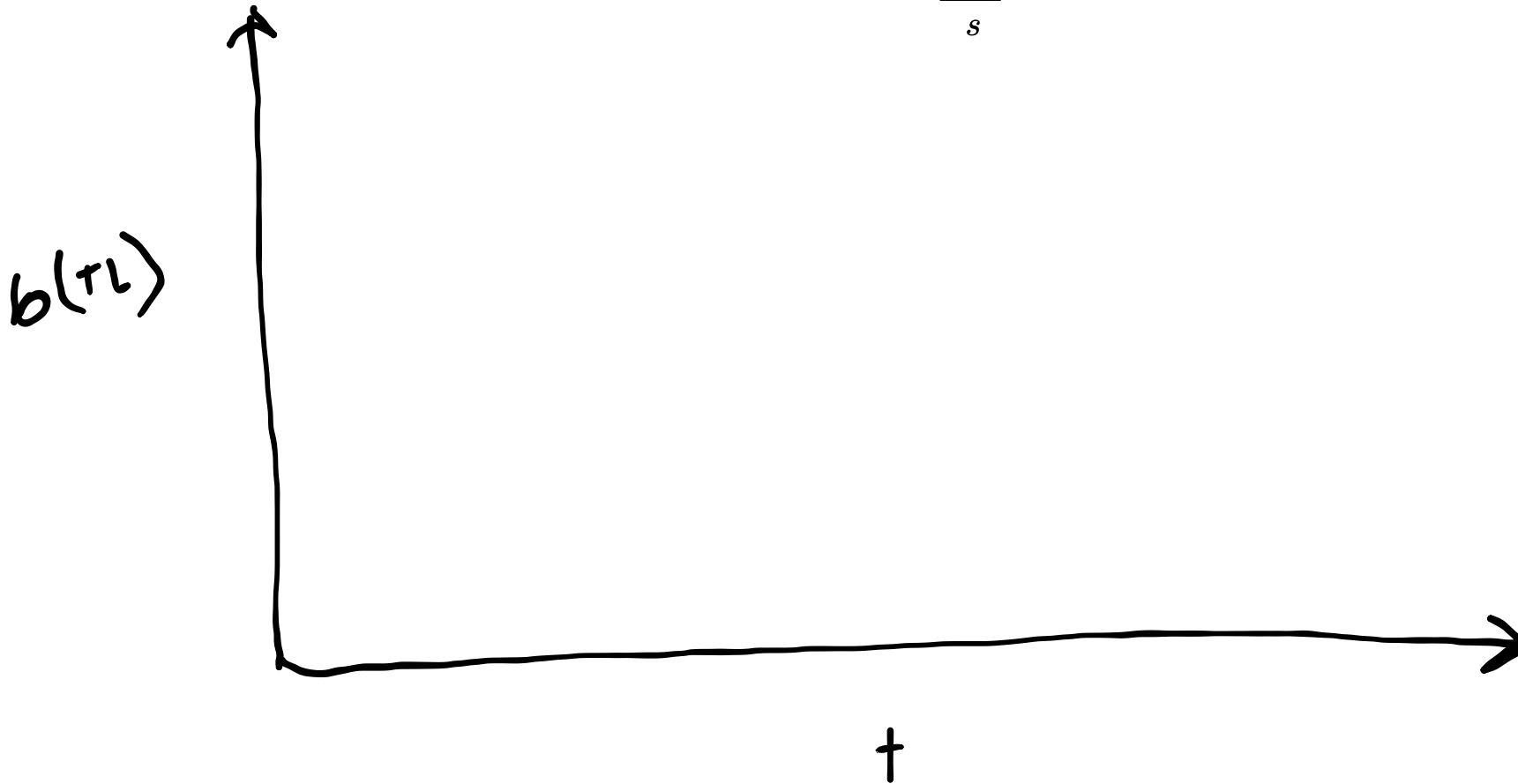
$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

$b(\tau_L)$

†

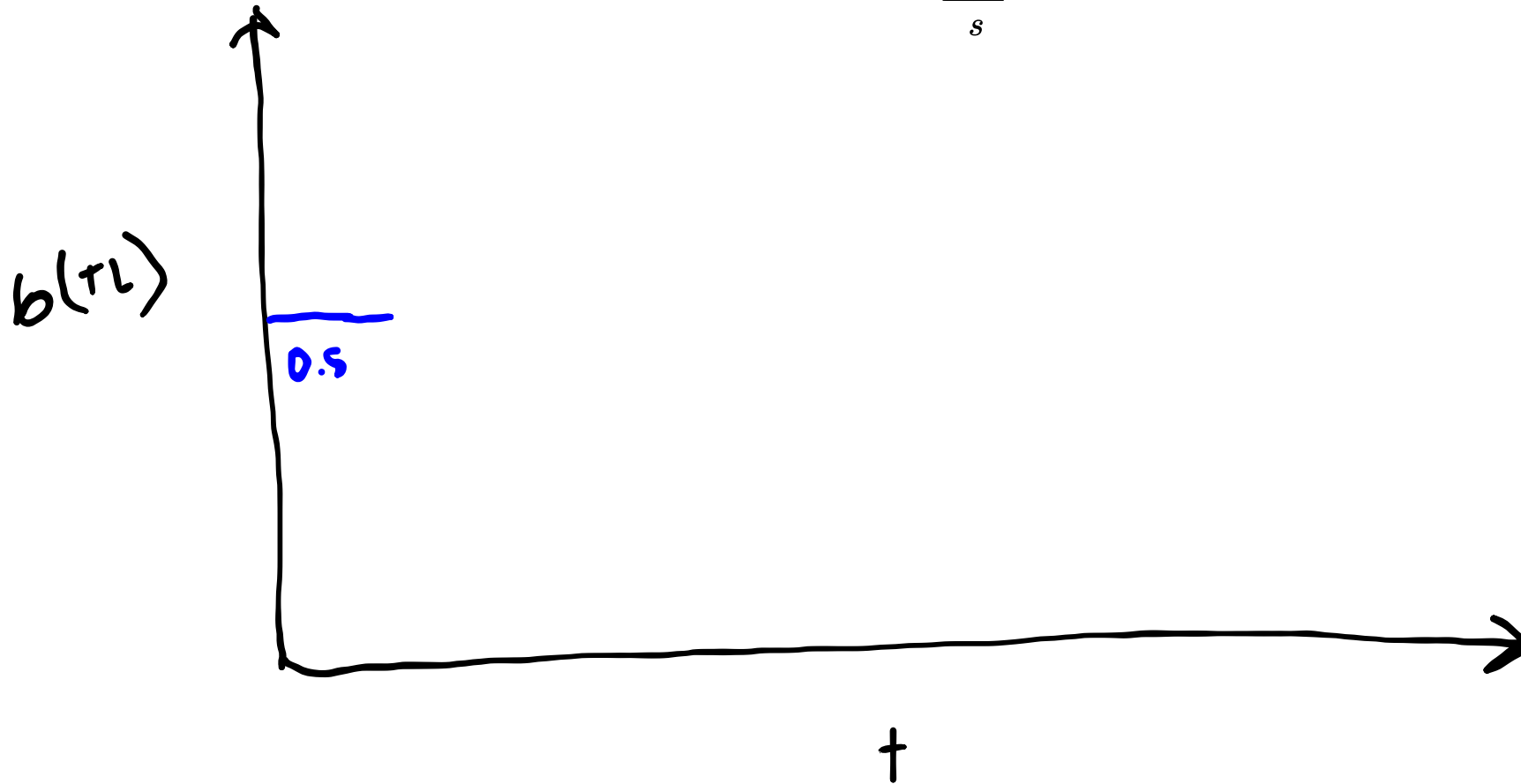
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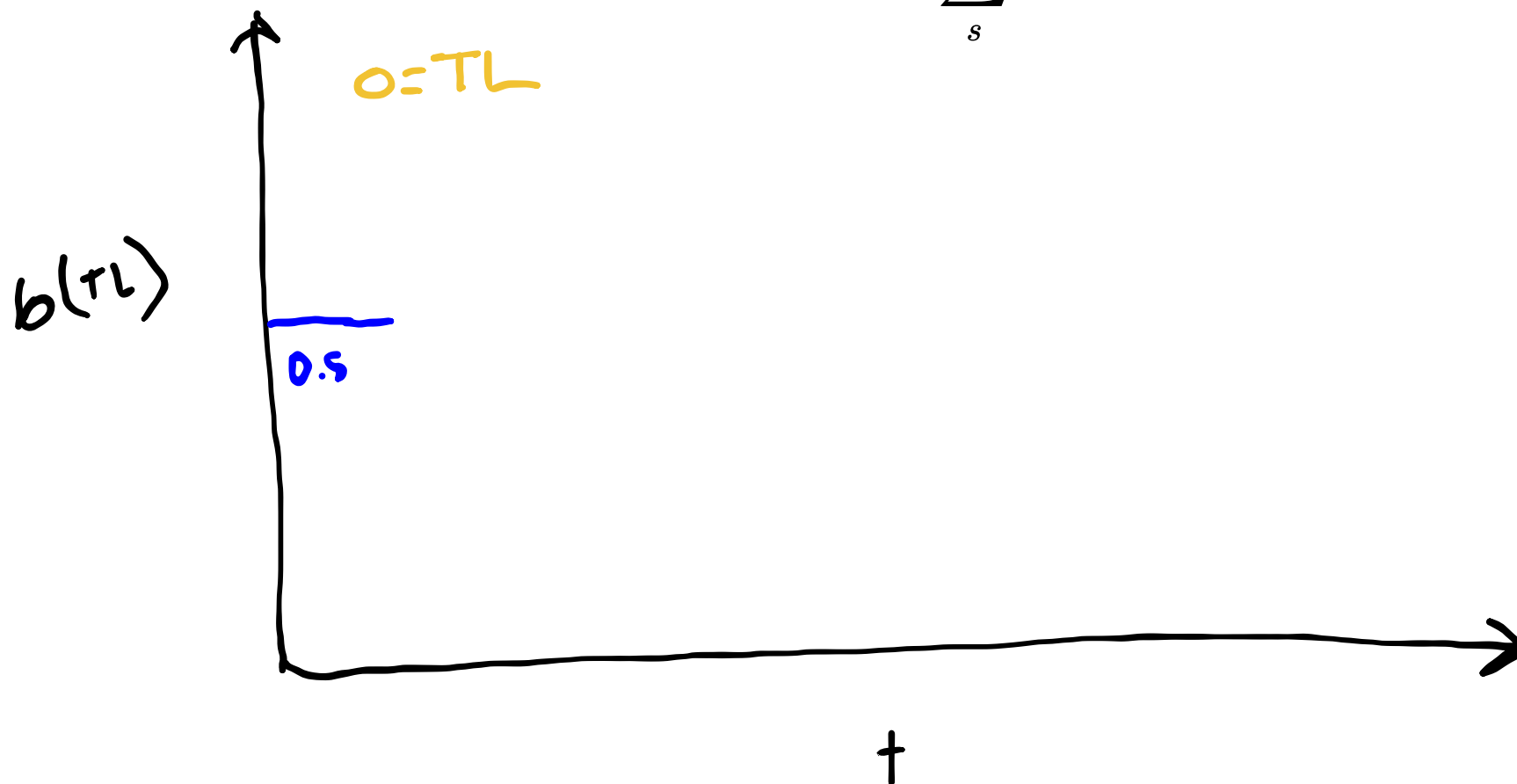
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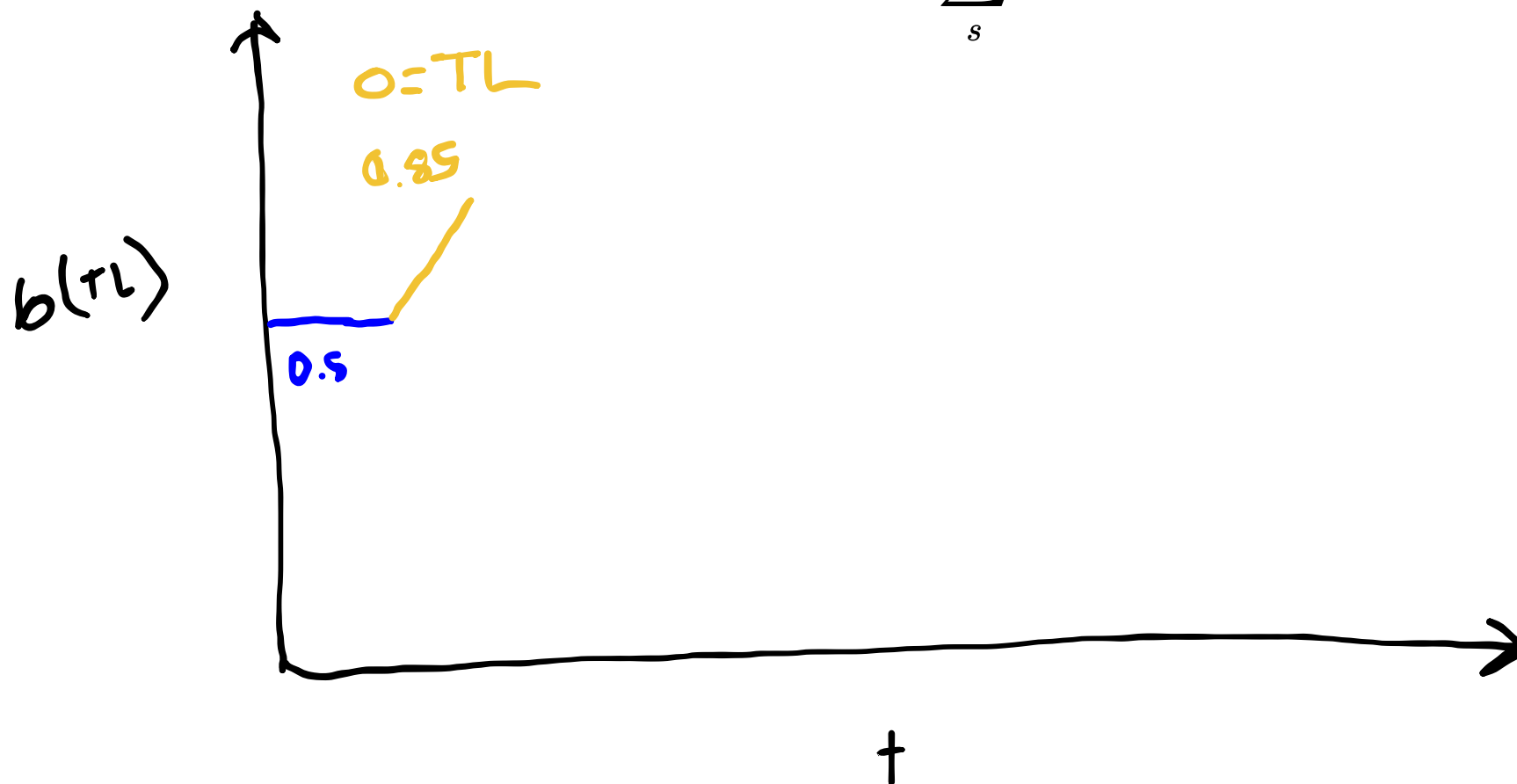
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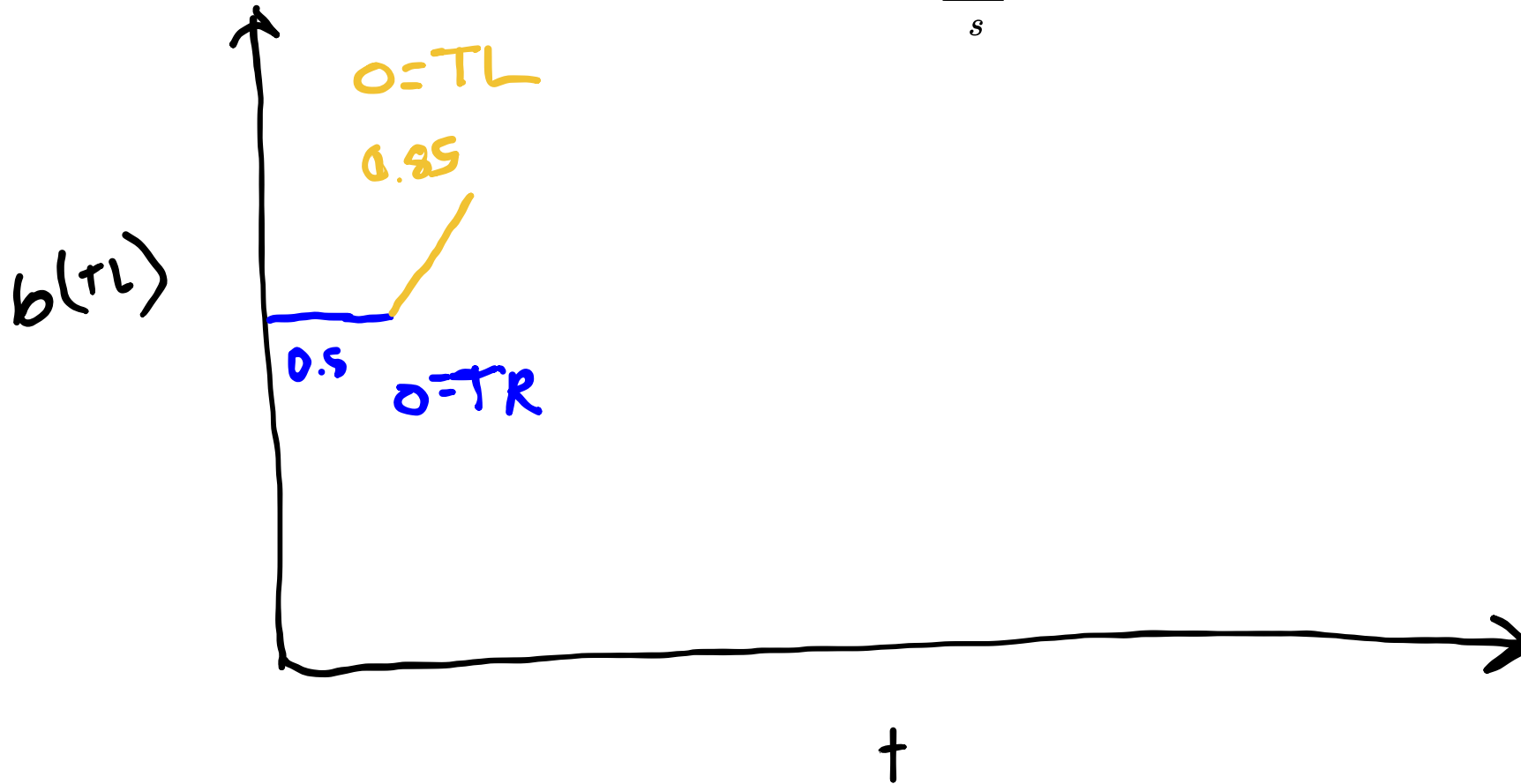
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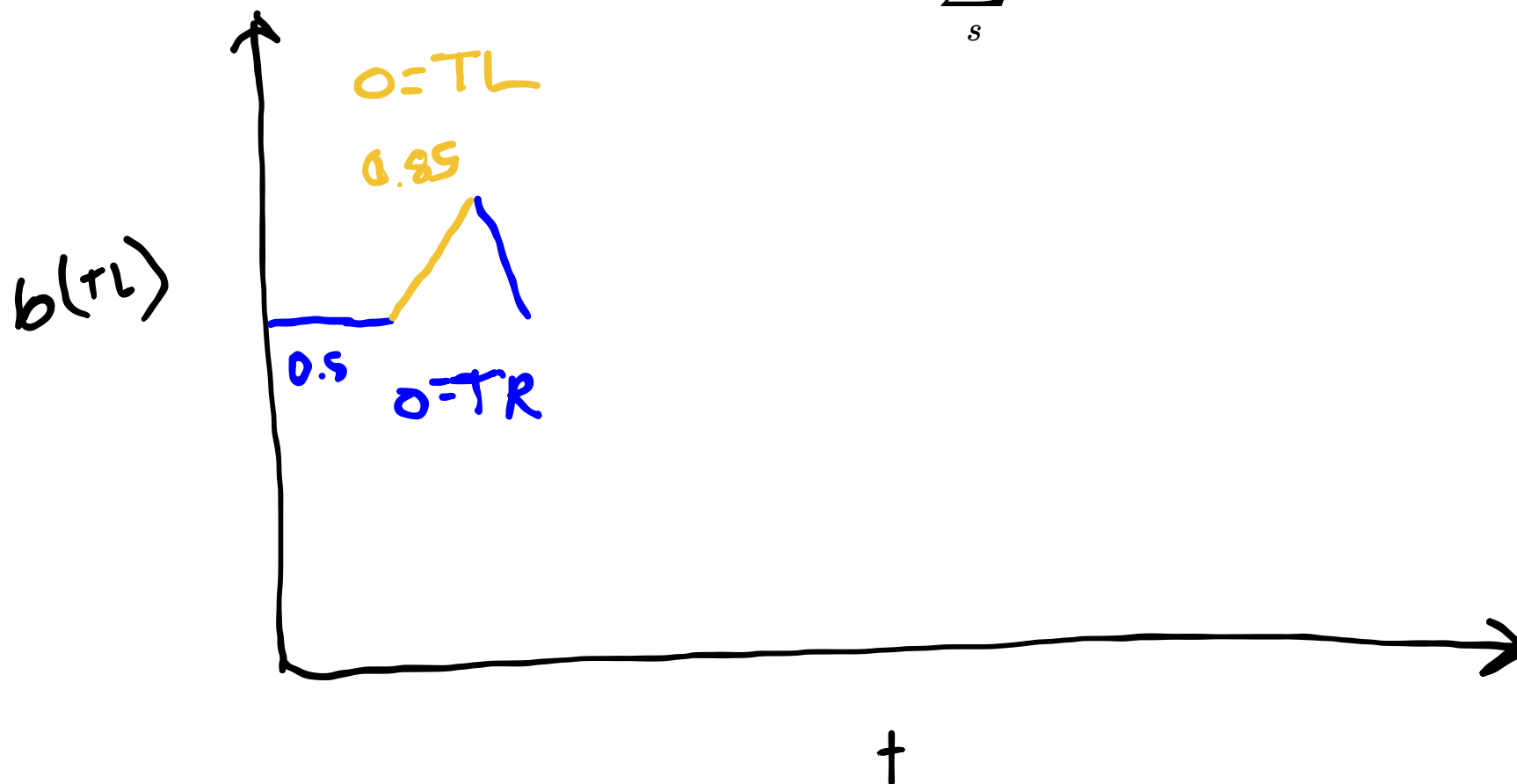
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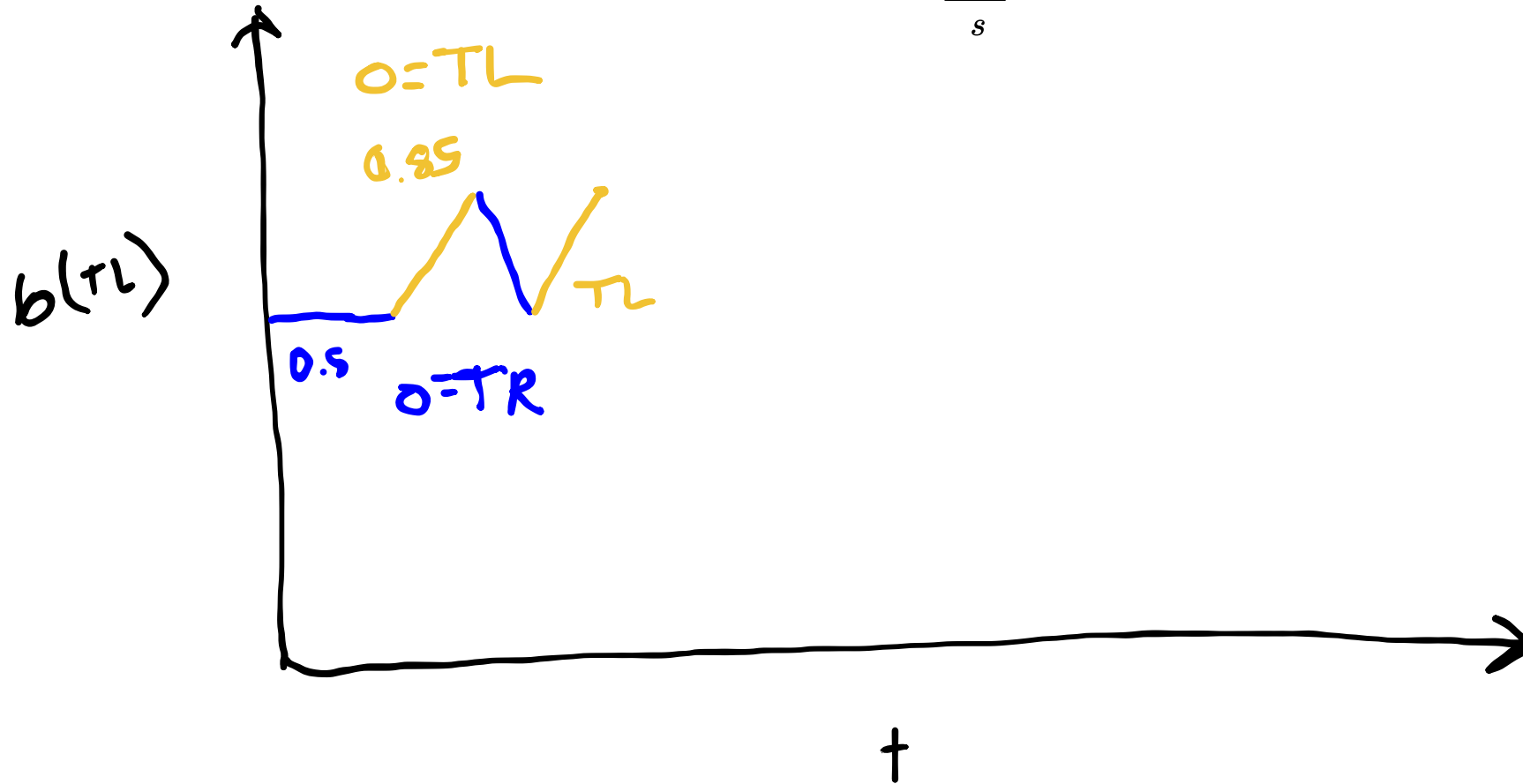
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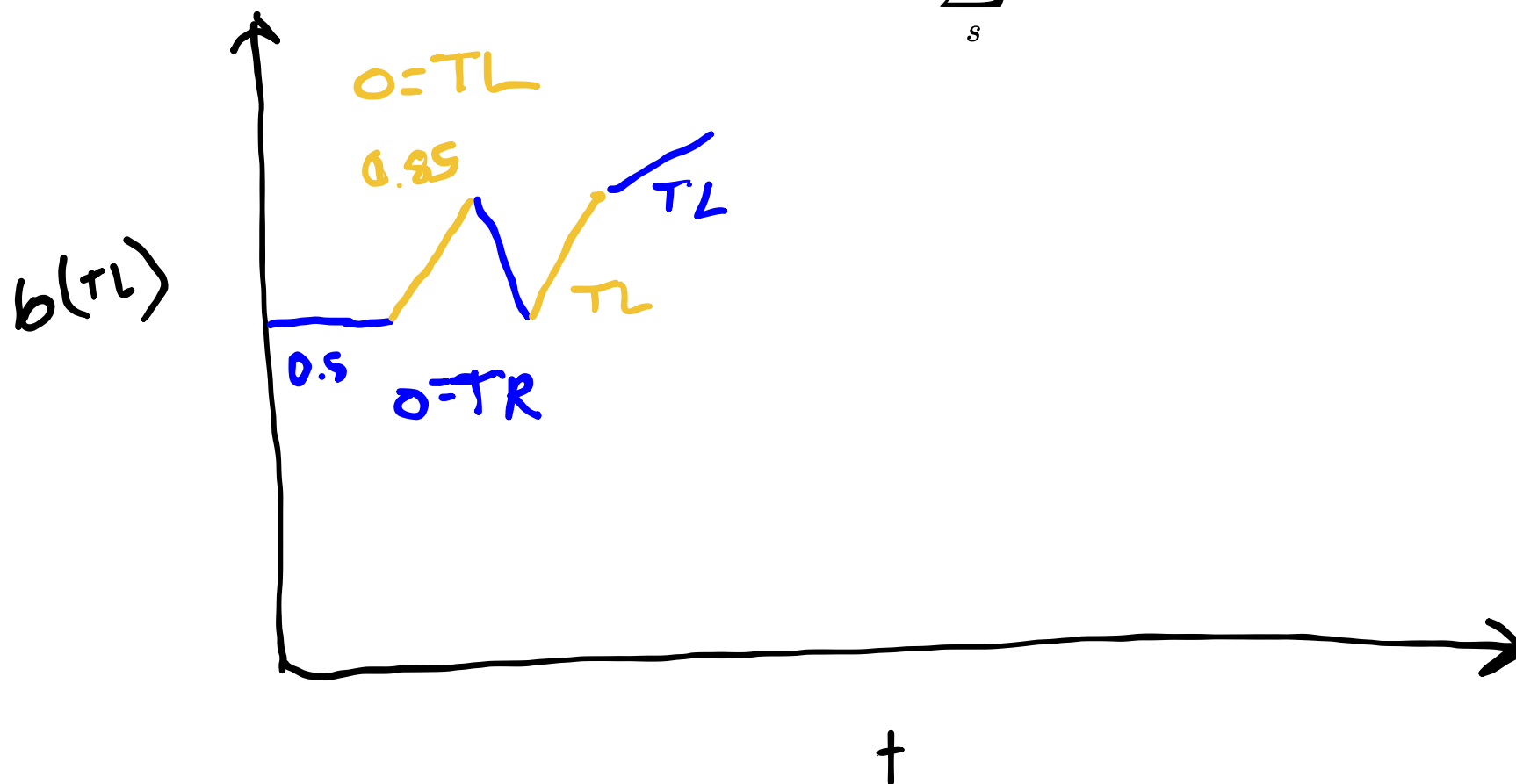
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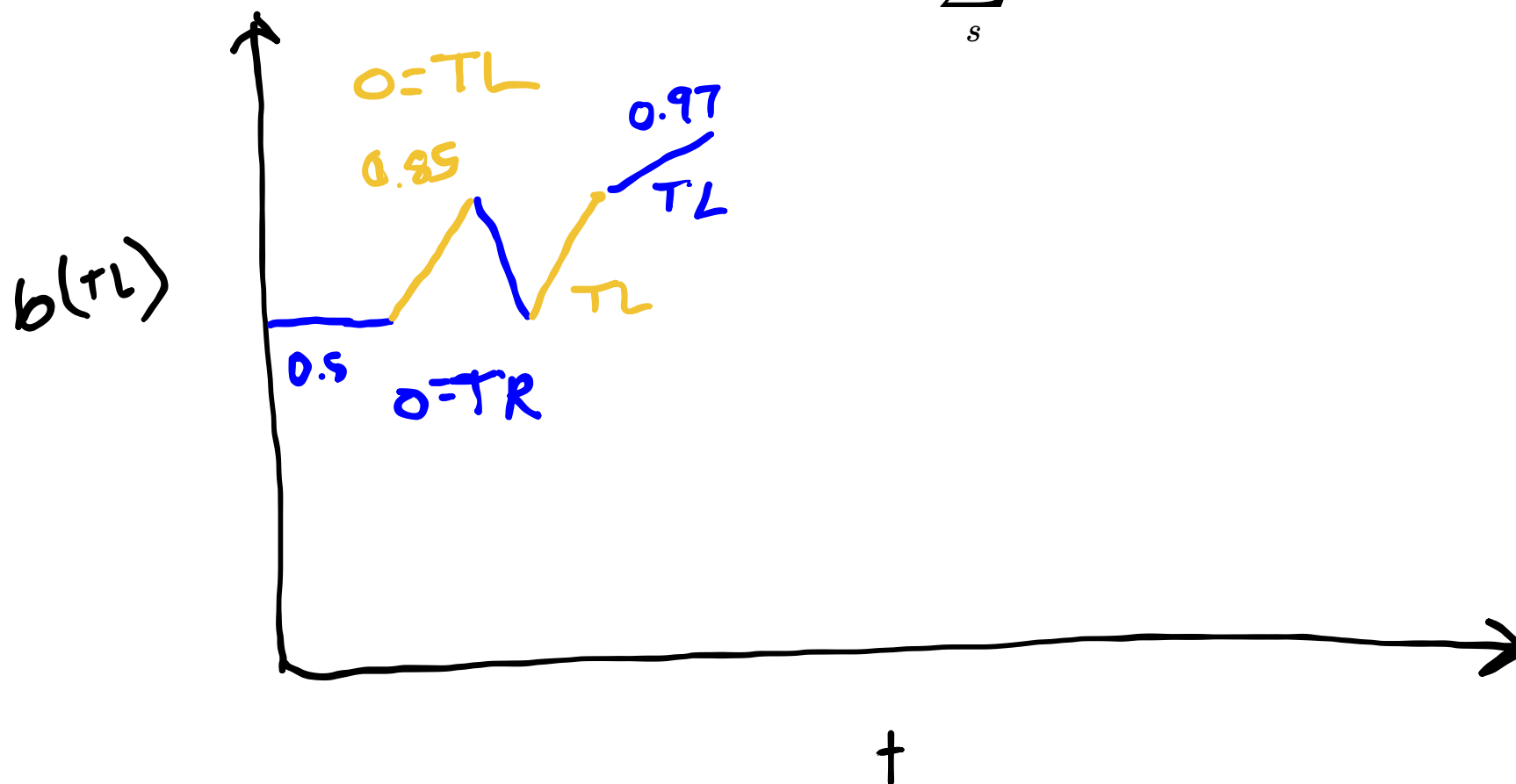
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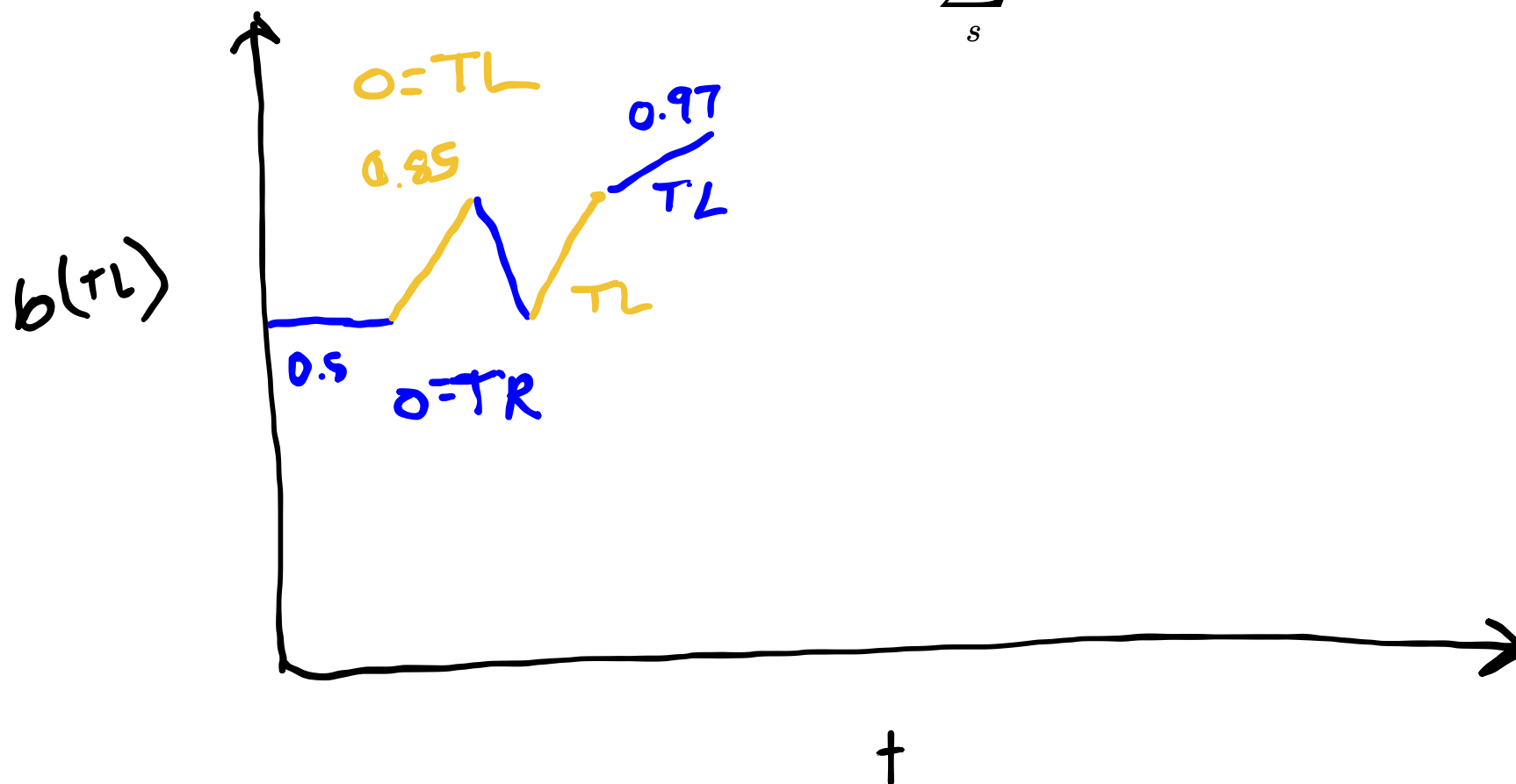
Belief Dynamics

$$b'(s') \propto Z(o | a, s') \sum_s T(s' | s, a) b(s)$$



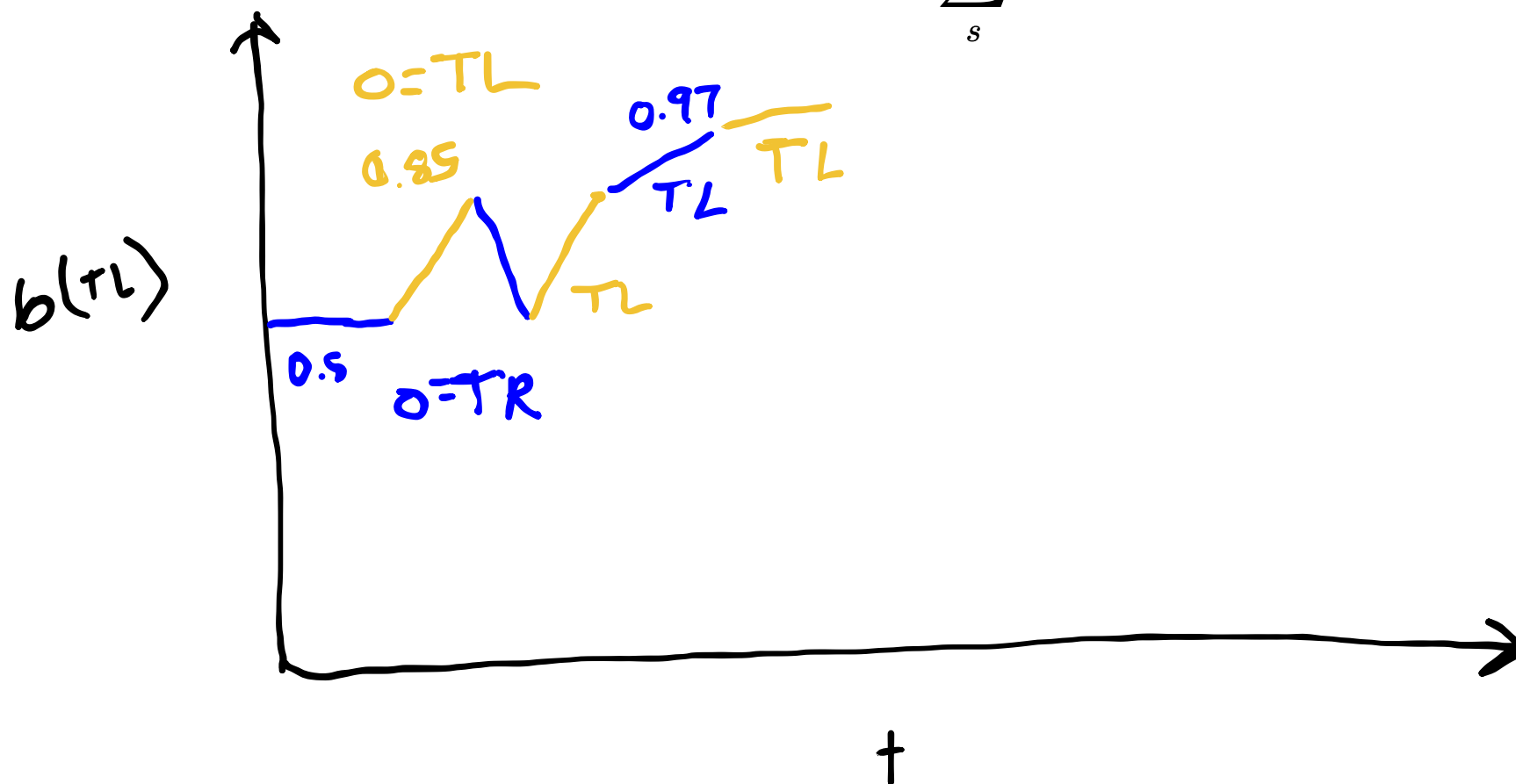
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Reward

- +10 opening empty
- 1 listening ←
- 100 opening tiger door
- 0 do nothing

One-step utility

(reward)

Want
 $R(b, a)$

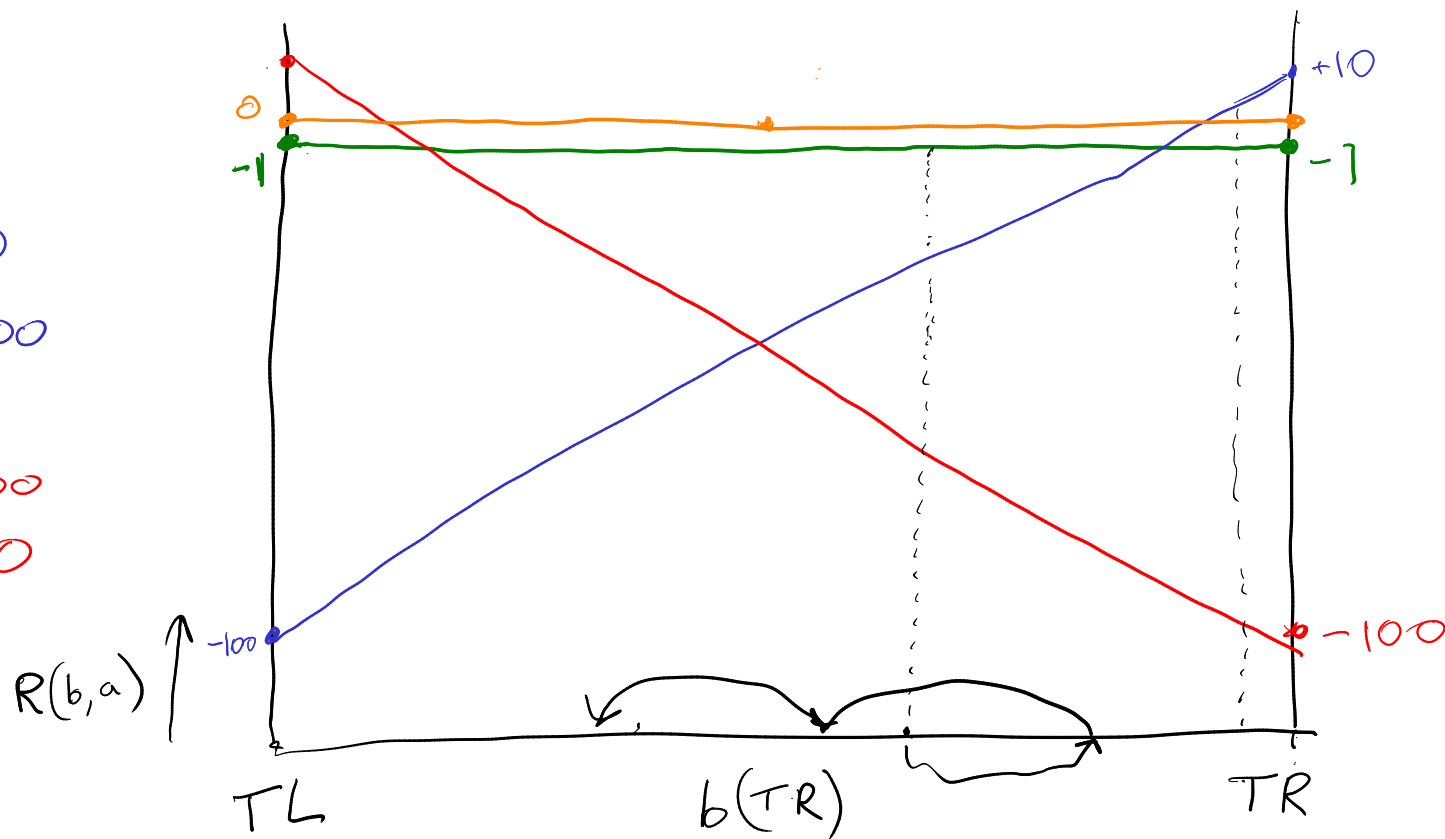
one-step
α-vector
for action α

$$R(b, a) = \vec{R}^a \cdot \vec{b}$$

$\downarrow$
 $\begin{bmatrix} R(s^1, a) \\ R(s^2, a) \\ \vdots \\ R(s^n, a) \end{bmatrix}$

$\downarrow$
 $\begin{bmatrix} b(s^1) \\ b(s^2) \\ \vdots \\ b(s^n) \end{bmatrix}$

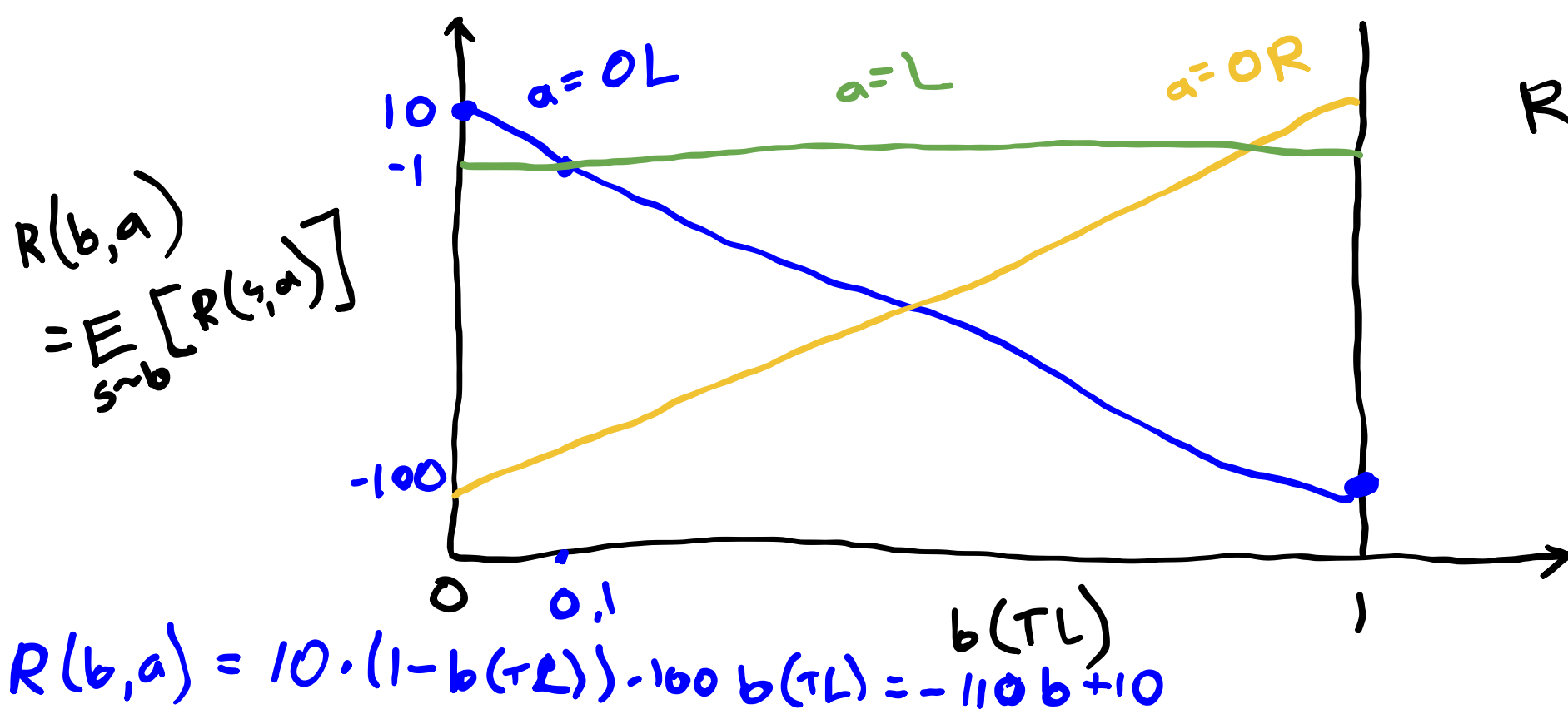
$$\pi_{\text{greedy}}(b) = \arg\max_a \vec{R}^a \cdot \vec{b}$$



$$R(b, a) = E[R(s, a)] = \sum_s b(s) R(s, a) = b(TL) R(TL, a) + (1 - b(TL)) R(TR, a)$$

One-step utility

Reward: +10 empty door
 -1 Listen
 -100 Tiger



$$R(b, a) = \bar{r}_a \cdot b$$

$\uparrow$
 α -vector

Exercise 2: Crying Baby 1-Step Utility

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

Draw the 1-step utility α -vectors
for the Crying Baby problem.

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

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$$\gamma = 0.9$$

Alpha Vectors for Conditional Plans

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Conditional Plans: fixed-depth history-based policies

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1 Step:

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (oL) (oR)

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

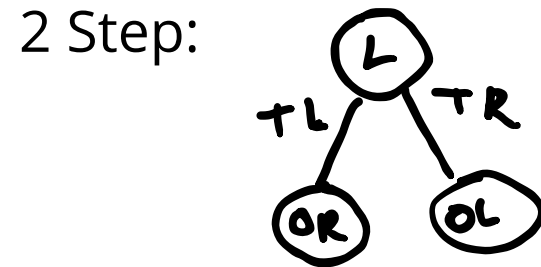
1 Step: (L) (oL) (oR)

2 Step:

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

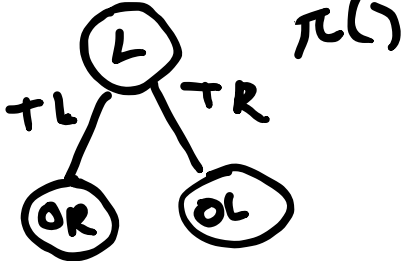
1 Step: (L) (OL) (OR)



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Conditional Plans: fixed-depth history-based policies

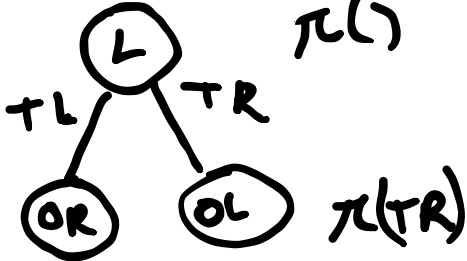
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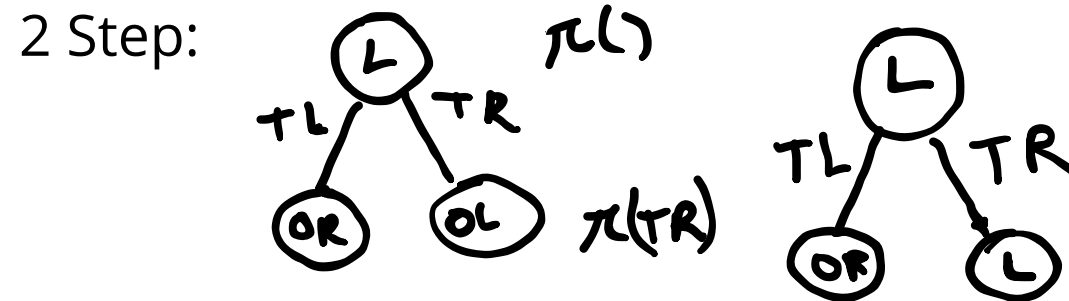
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2 Step: 

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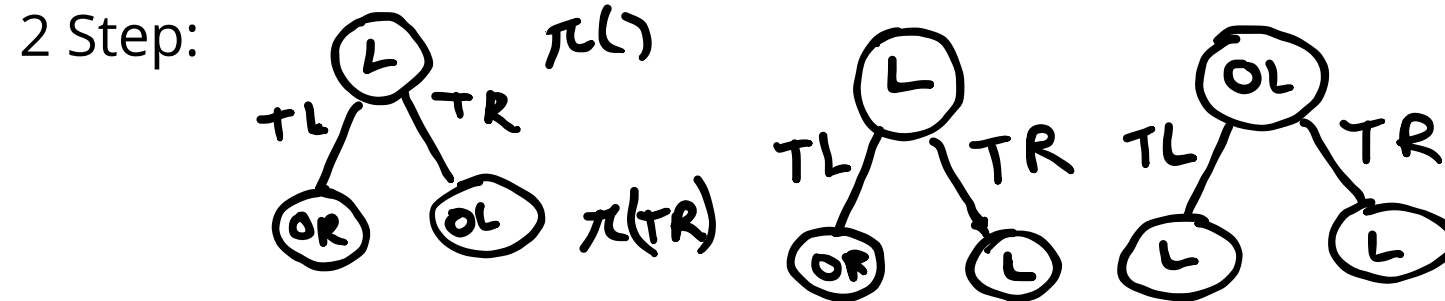
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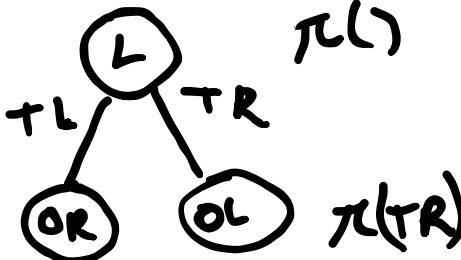
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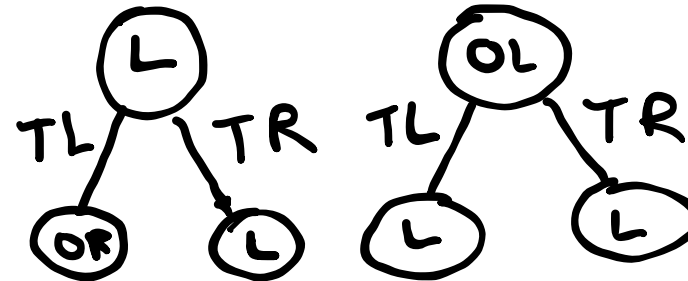


Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)

2 Step:  $\pi(L)$

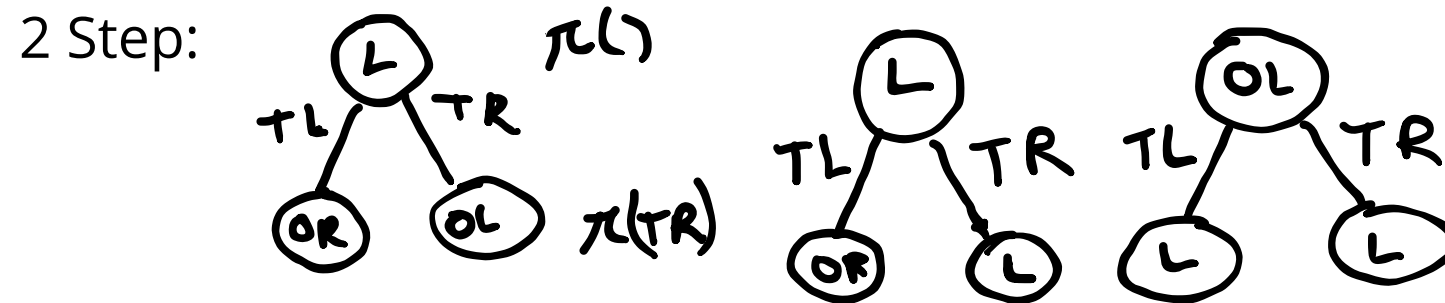


$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)



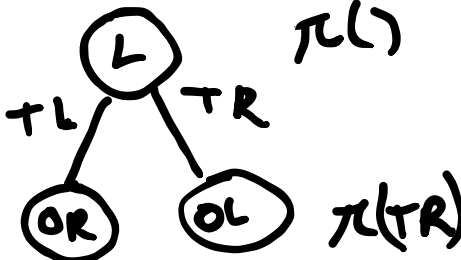
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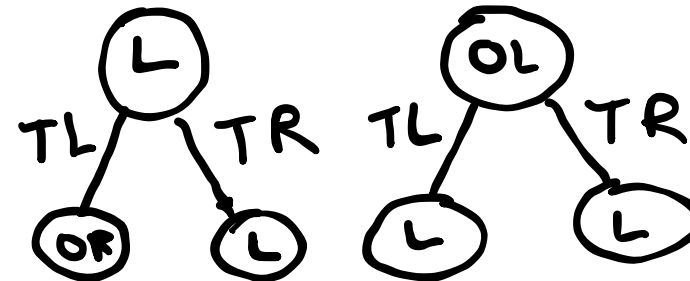
27 two step plans!

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

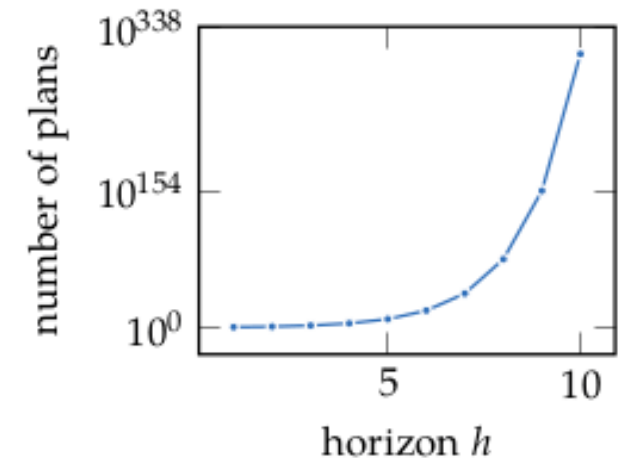
1 Step: (L) (OL) (OR)

2 Step:  $\pi(L)$ $\pi(TR)$



$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

27 two step plans!



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For 1-step: $U^\pi(s) = R(s, \pi())$

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$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

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Alpha Vectors for Conditional Plans

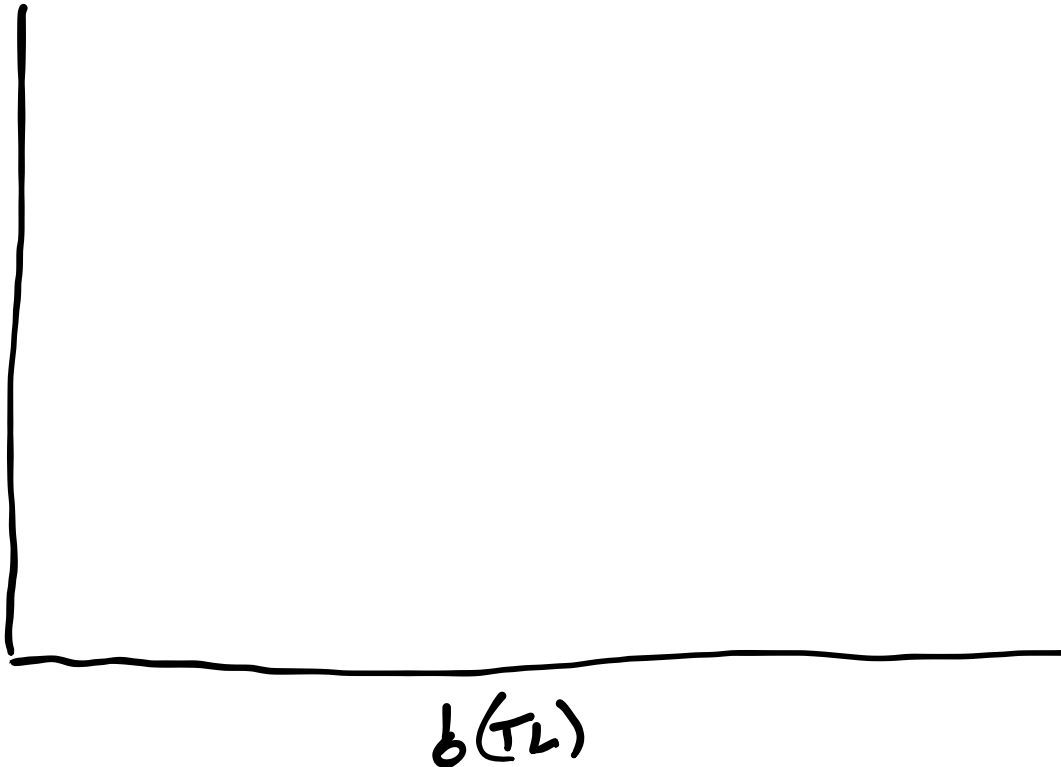
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For 1-step: $U^\pi(s) = R(s, \pi())$

Alpha Vectors for Conditional Plans

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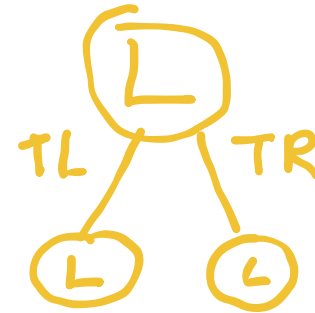
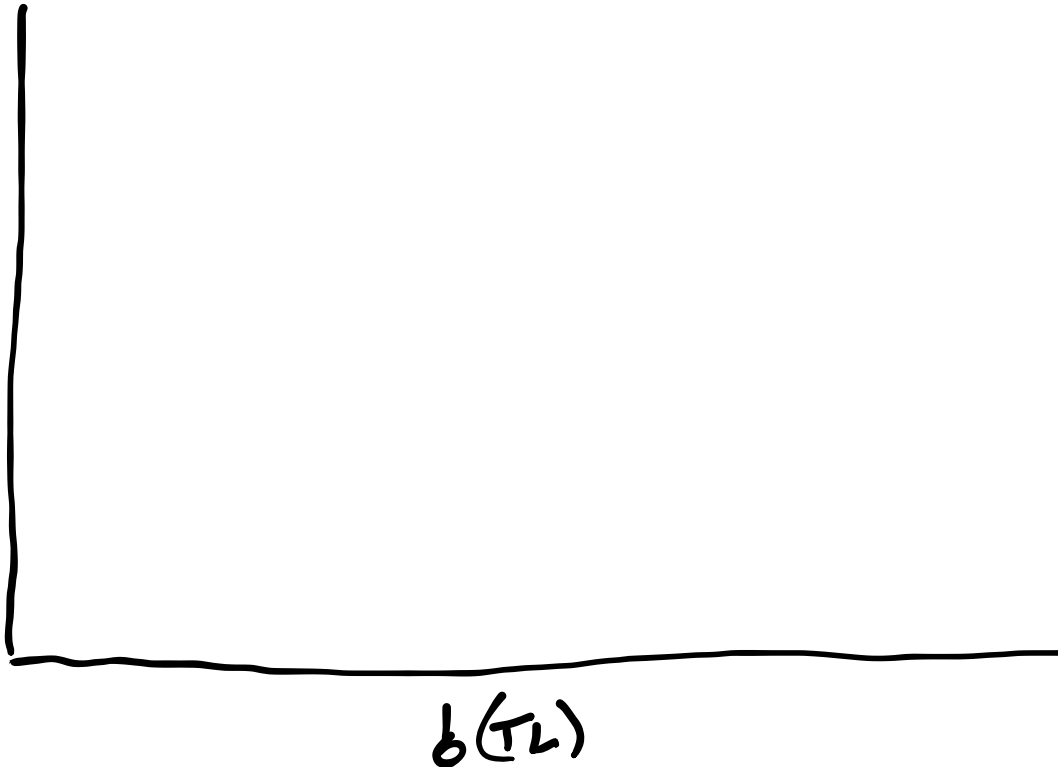
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Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

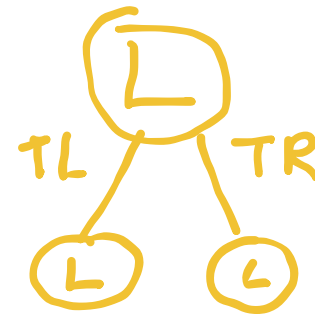
For 1-step: $U^\pi(s) = R(s, \pi())$



Alpha Vectors for Conditional Plans

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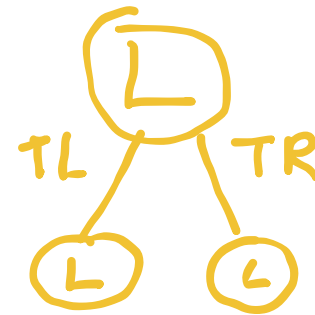
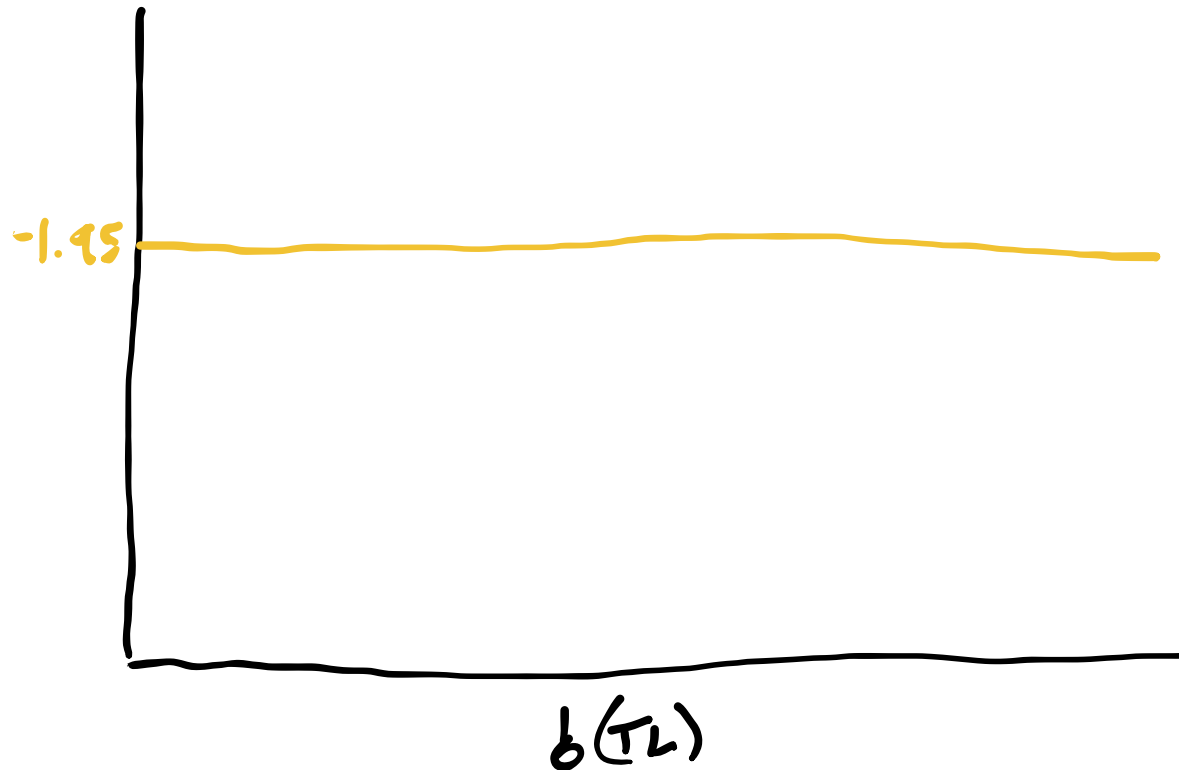
$$U^\pi(s) = -1 + \gamma(-1)$$

$b(\pi_L)$

Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

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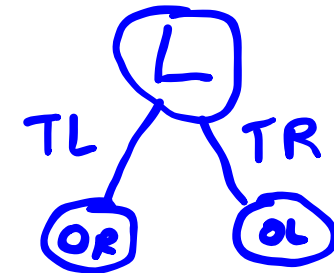
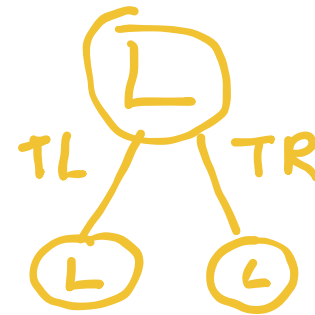
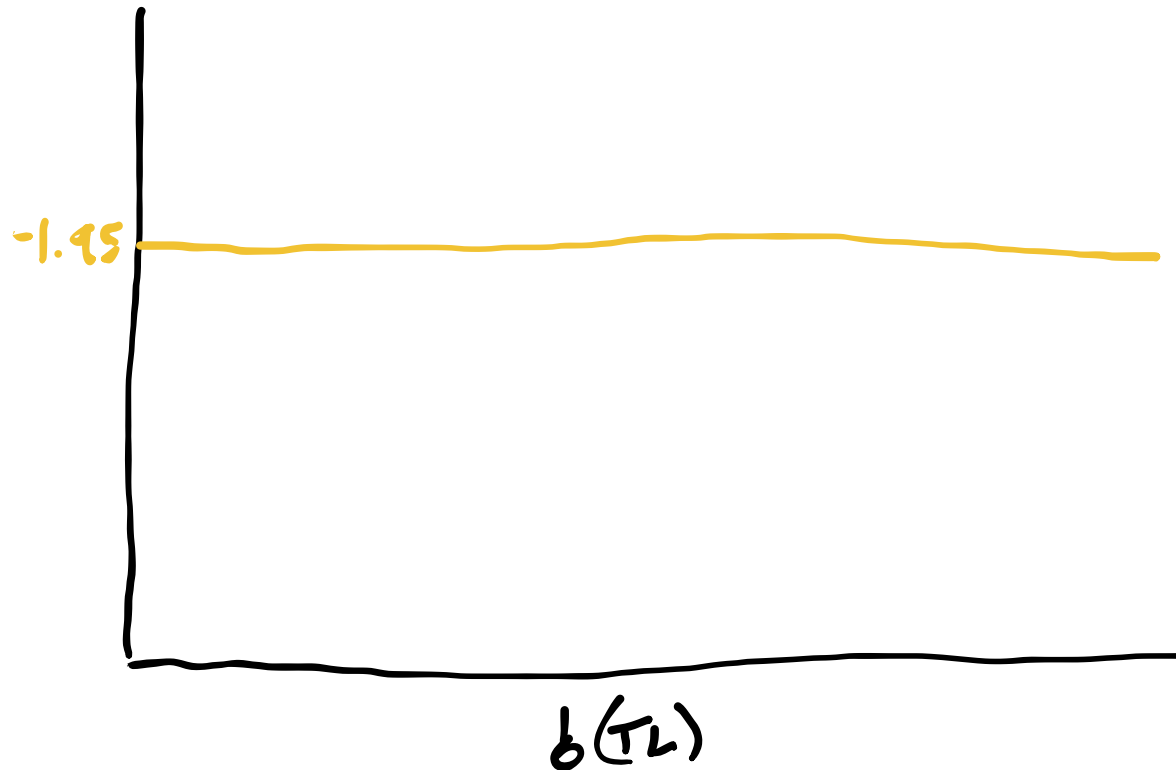


$$U^\pi(s) = -1 + \gamma(-1)$$

Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

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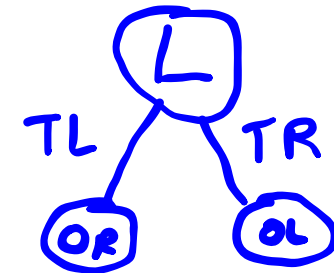
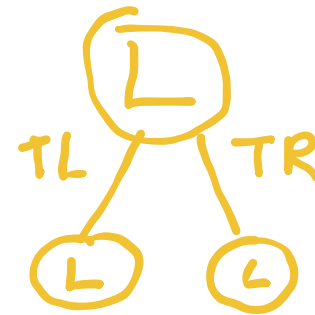
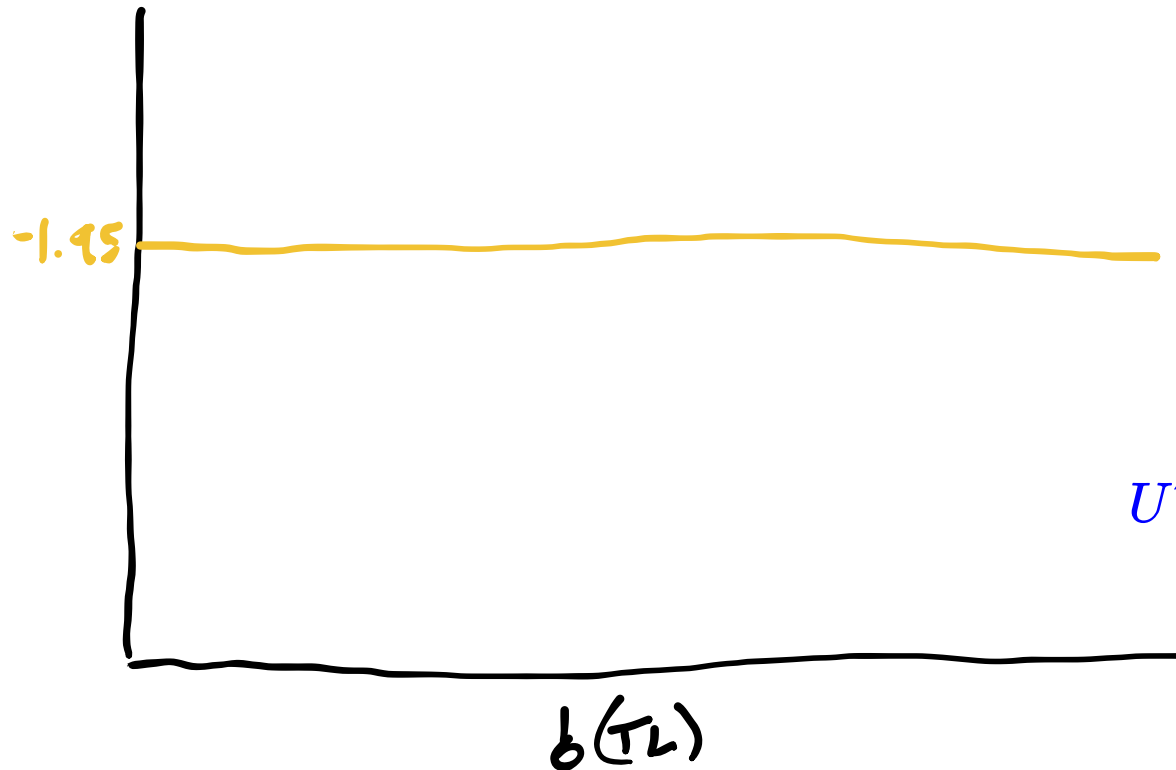


$$U^\pi(s) = -1 + \gamma(-1)$$

Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

For 1-step: $U^\pi(s) = R(s, \pi())$



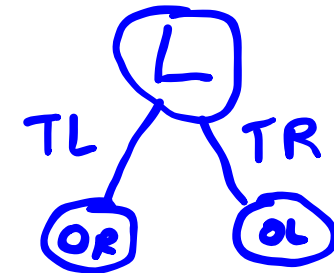
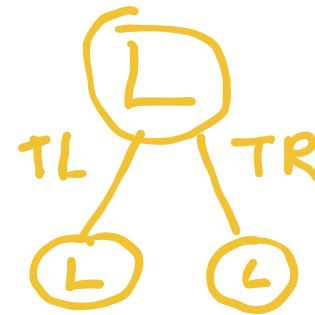
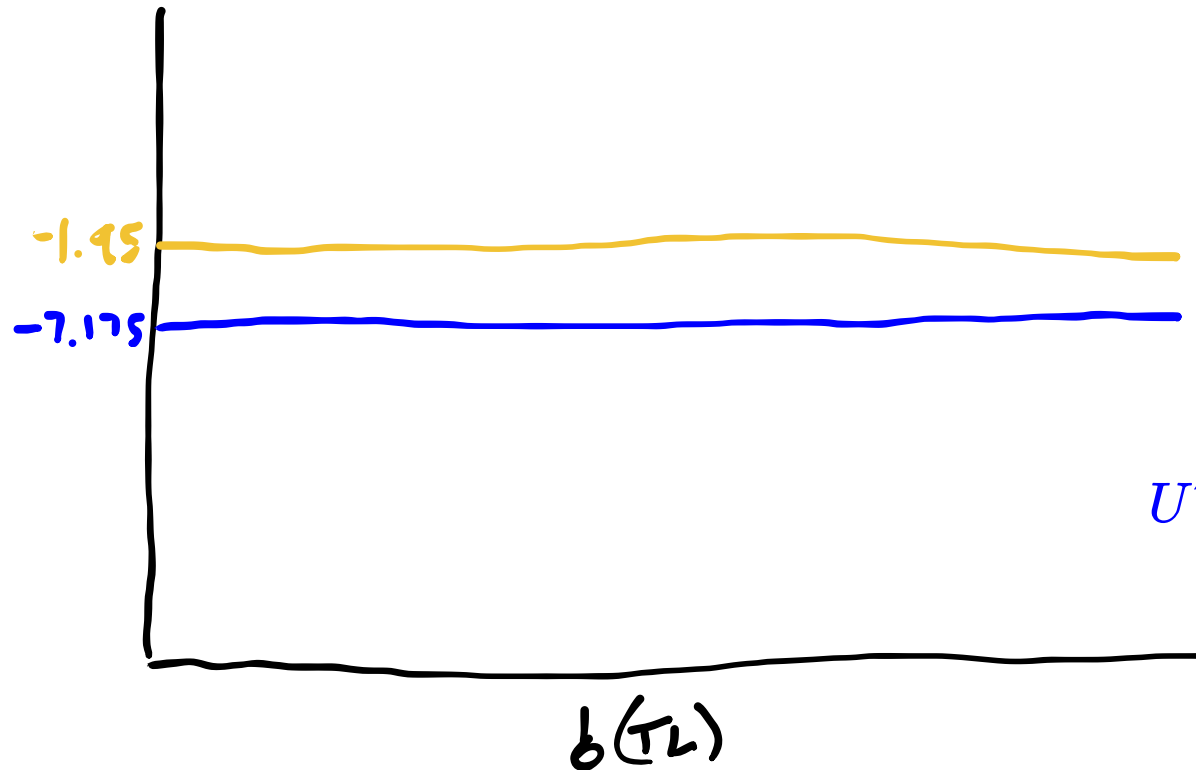
$$U^\pi(s) = -1 + \gamma(-1)$$

$$U^\pi(TL) = -1 + 0.95(1(0.85 \times 10 + 0.15 \times -100))$$

Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

For 1-step: $U^\pi(s) = R(s, \pi())$



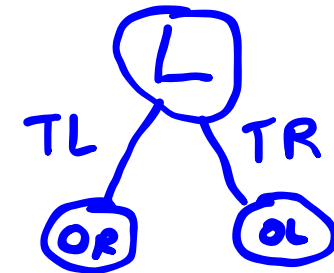
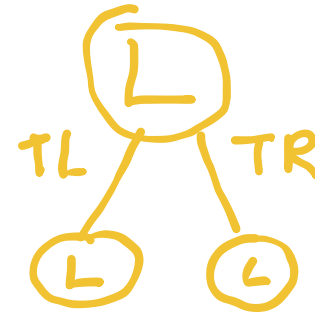
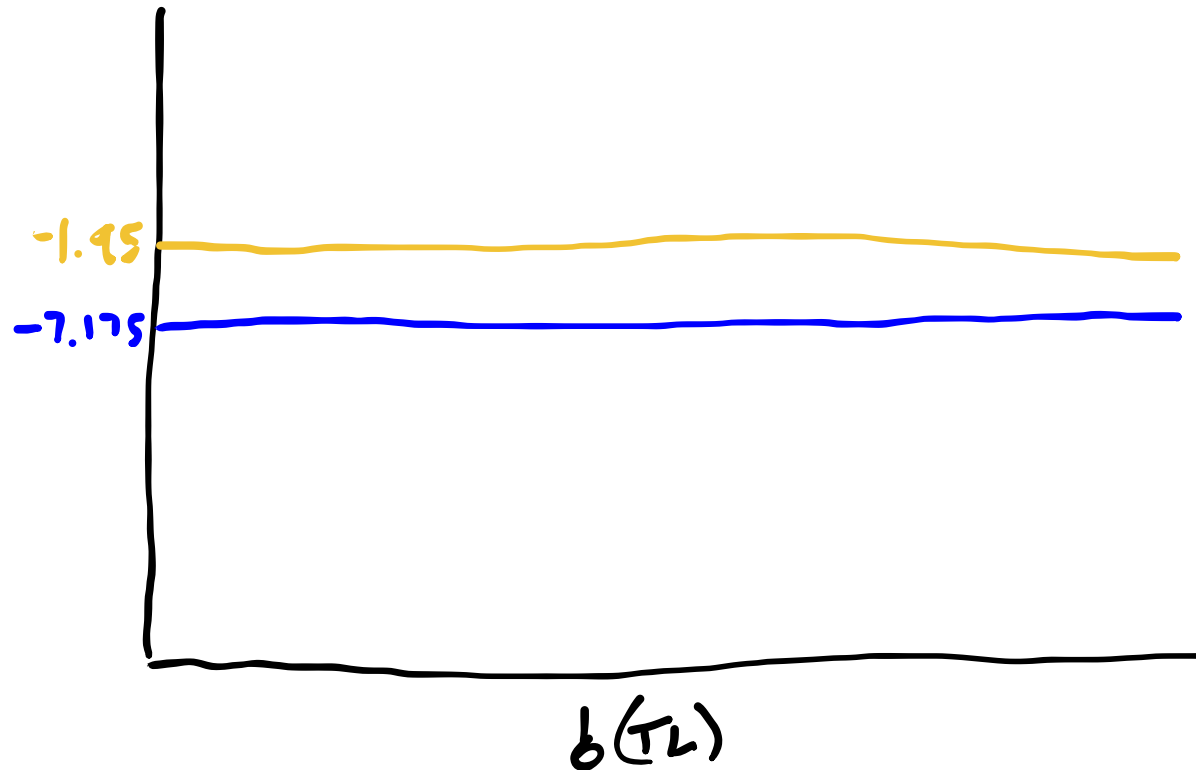
$$U^\pi(s) = -1 + \gamma(-1)$$

$$\begin{aligned} U^\pi(TL) &= -1 + 0.95(1(0.85 \times 10 + 0.15 \times -100)) \\ &= -7.175 \end{aligned}$$

Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

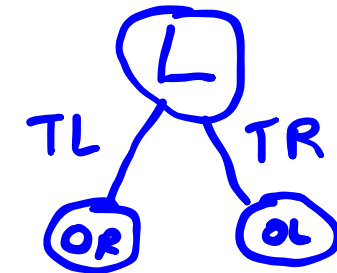
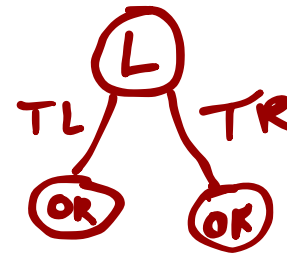
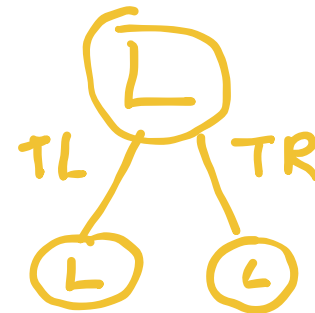
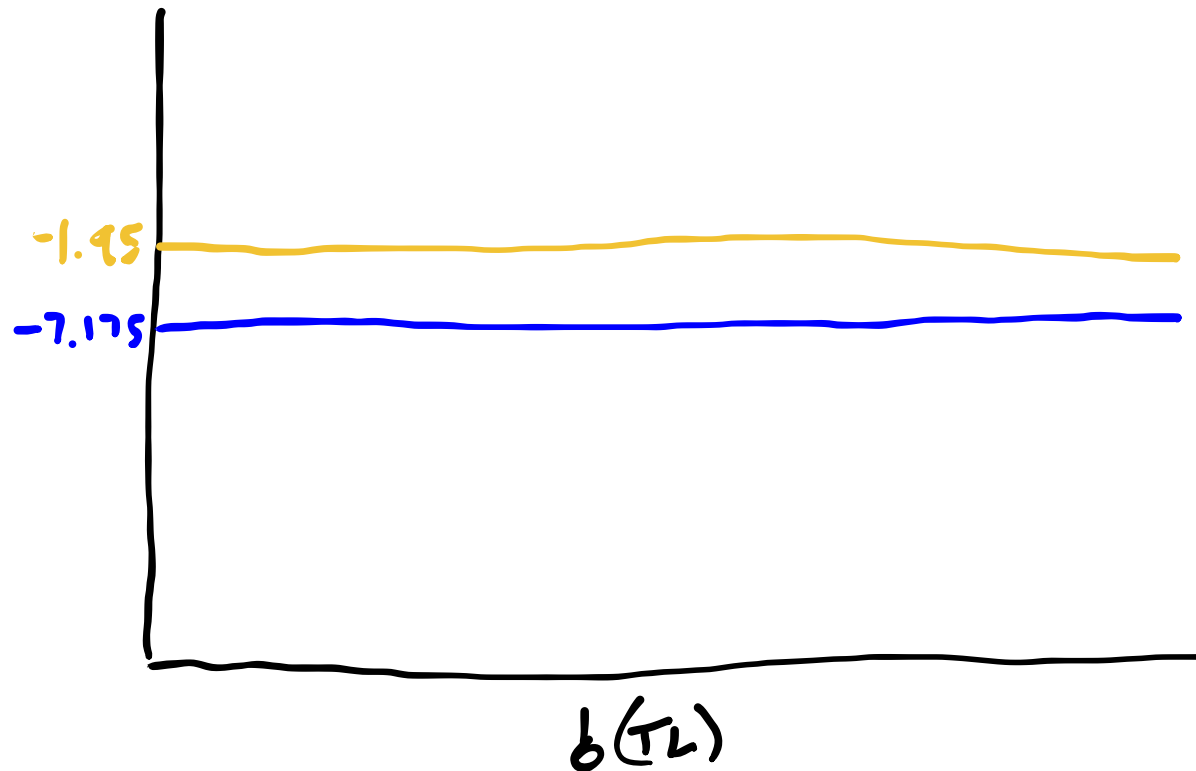
For 1-step: $U^\pi(s) = R(s, \pi())$



Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

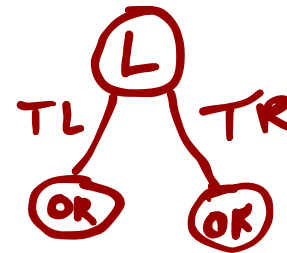
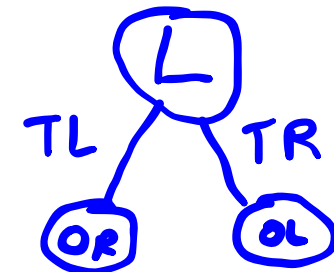
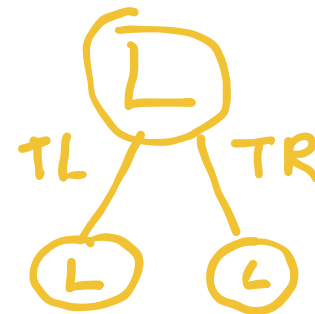
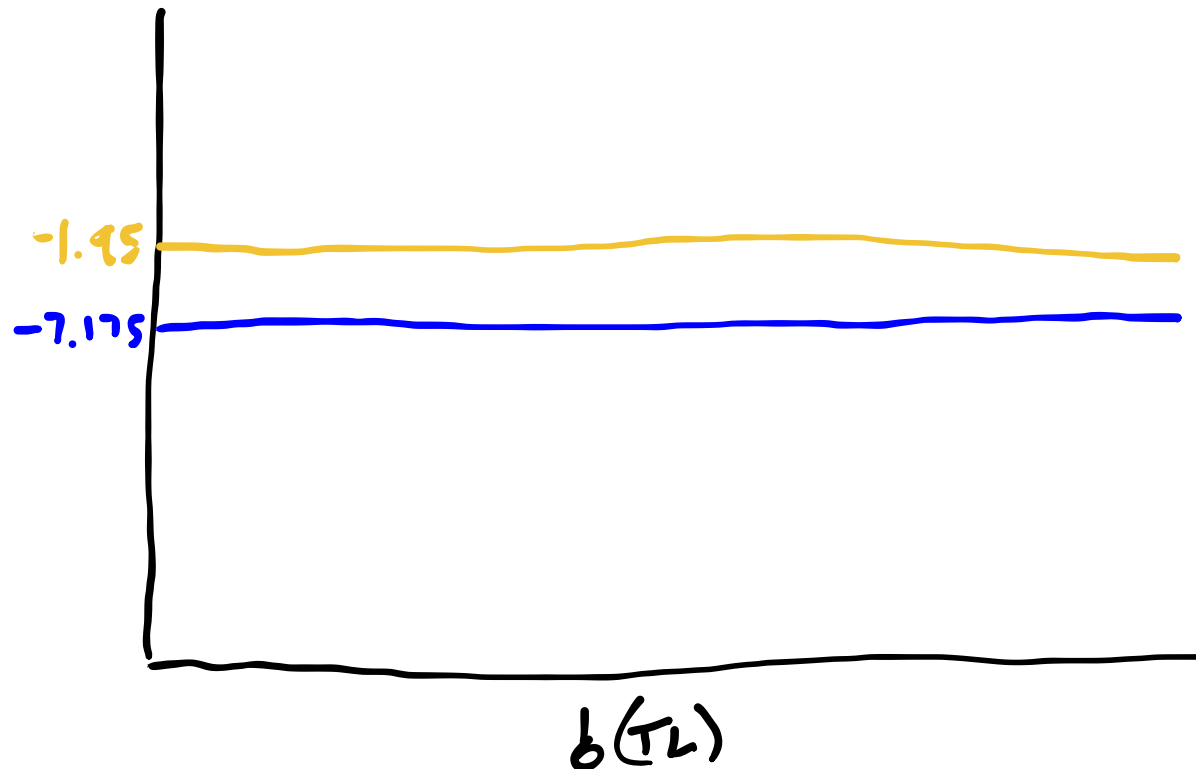
For 1-step: $U^\pi(s) = R(s, \pi())$



Alpha Vectors for Conditional Plans

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For 1-step: $U^\pi(s) = R(s, \pi())$

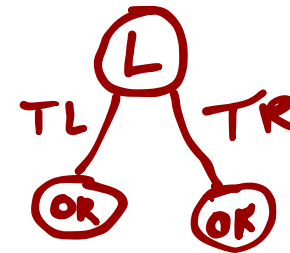
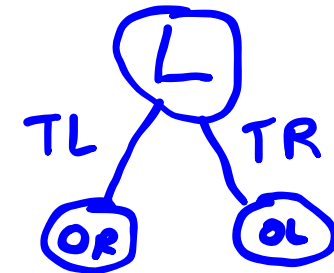
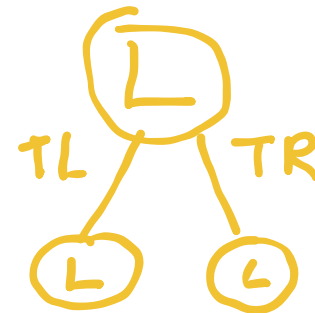
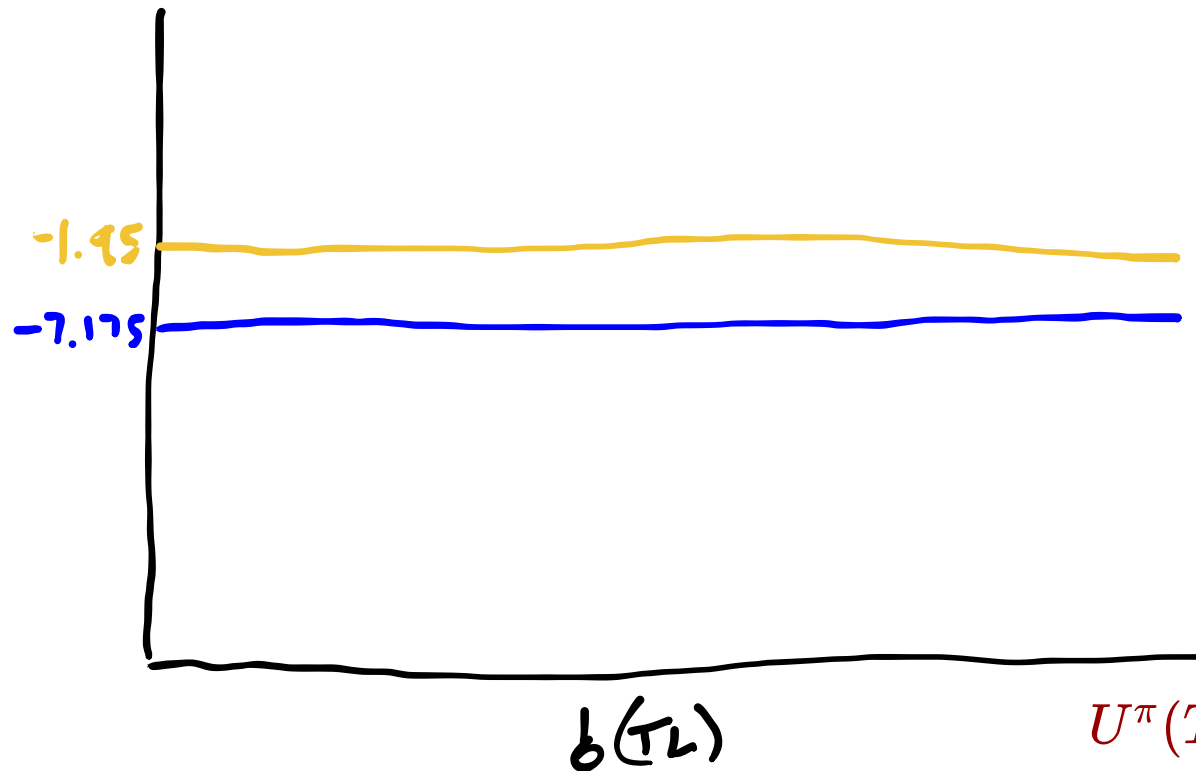


$$U^\pi(TL) = -1 + \gamma 10$$

Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

For 1-step: $U^\pi(s) = R(s, \pi())$



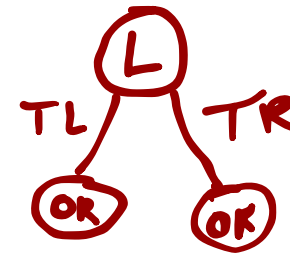
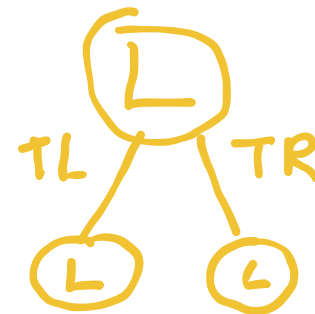
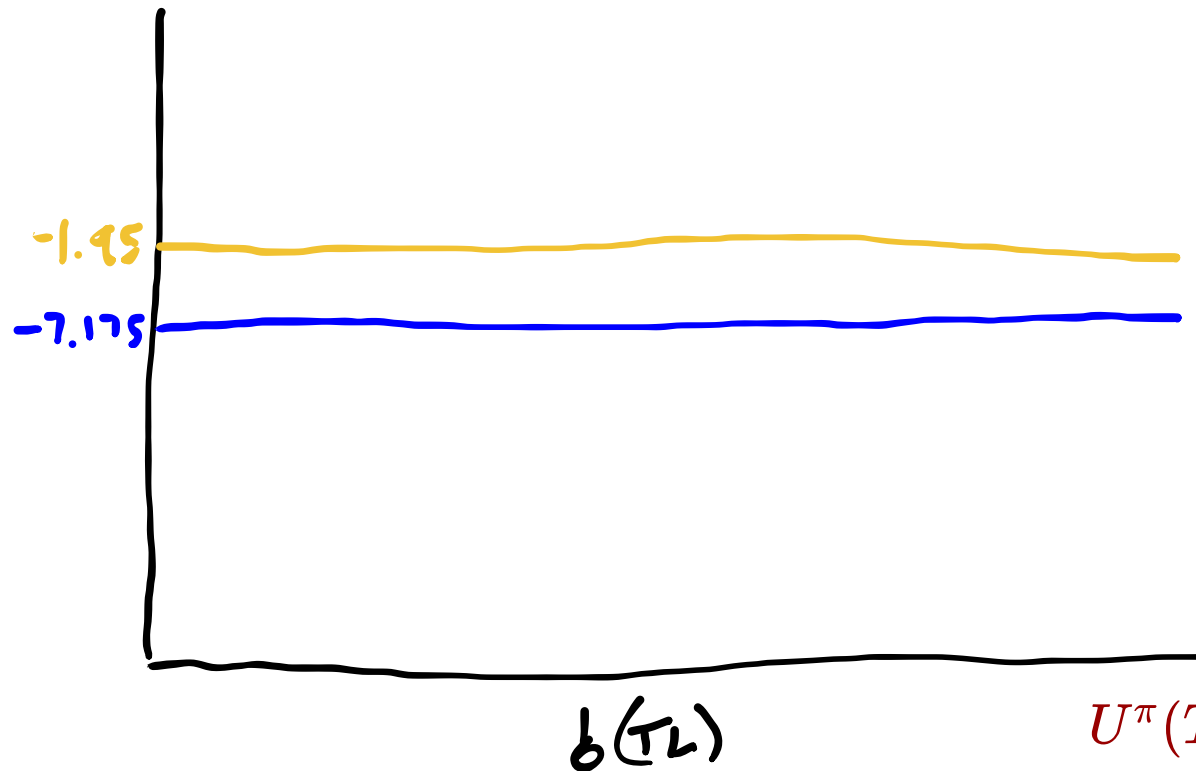
$$U^\pi(TL) = -1 + \gamma 10$$

$$U^\pi(TR) = -1 + \gamma(-100)$$

Alpha Vectors for Conditional Plans

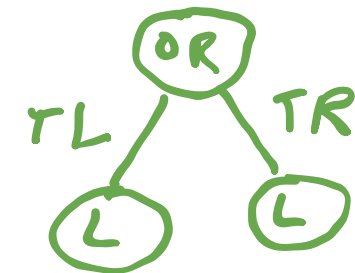
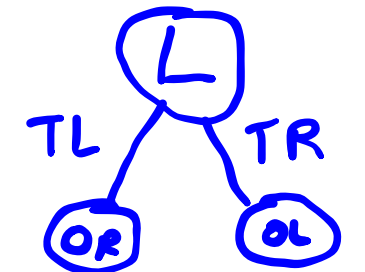
$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

For 1-step: $U^\pi(s) = R(s, \pi())$



$$U^\pi(TL) = -1 + \gamma 10$$

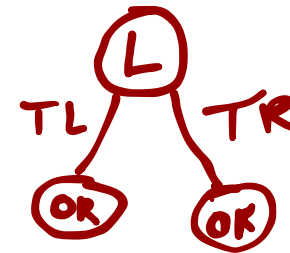
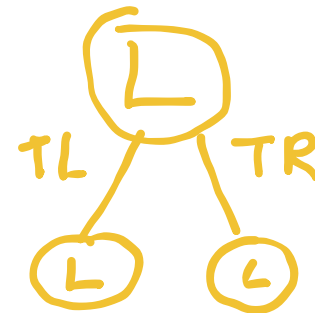
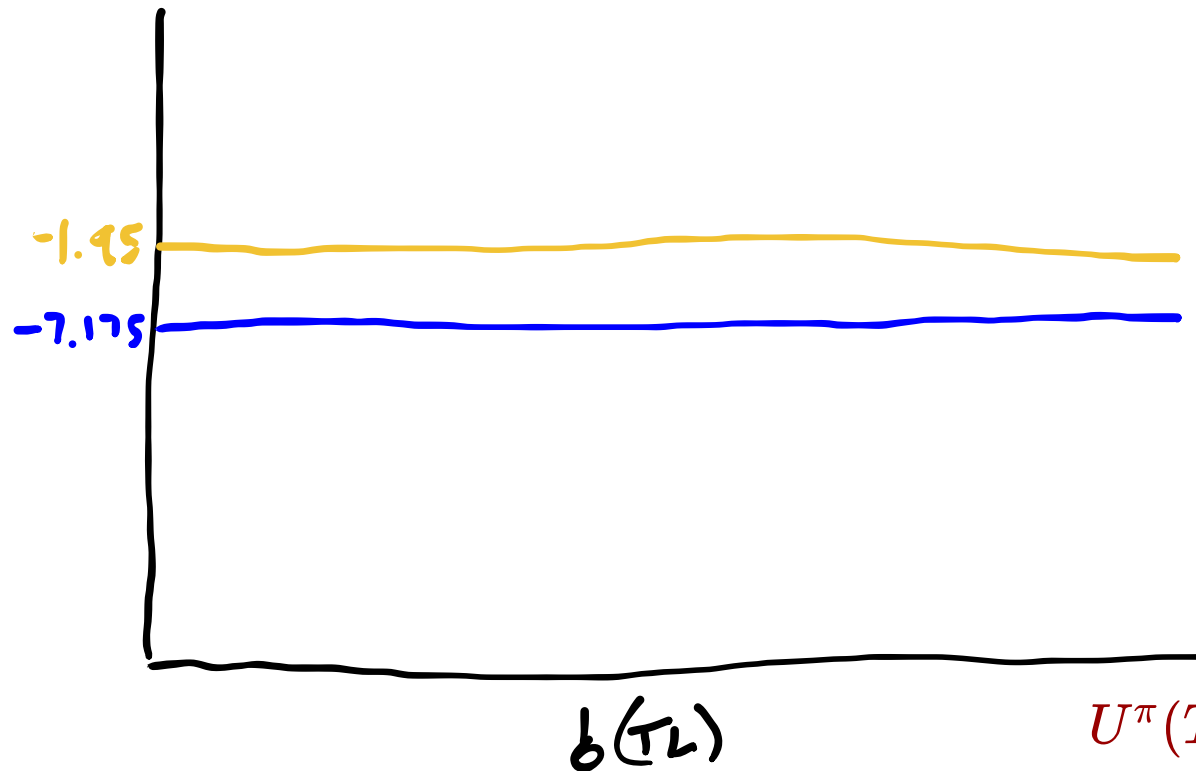
$$U^\pi(TR) = -1 + \gamma(-100)$$



Alpha Vectors for Conditional Plans

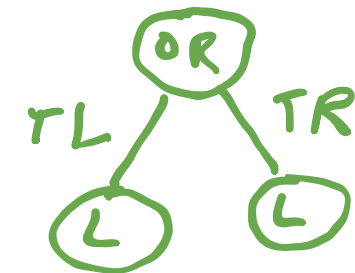
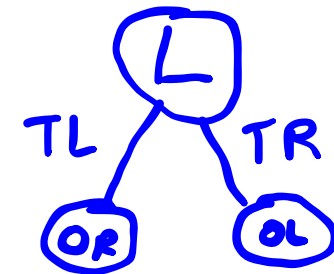
$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

For 1-step: $U^\pi(s) = R(s, \pi())$



$$U^\pi(TL) = -1 + \gamma 10$$

$$U^\pi(TR) = -1 + \gamma(-100)$$



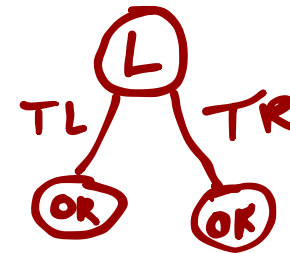
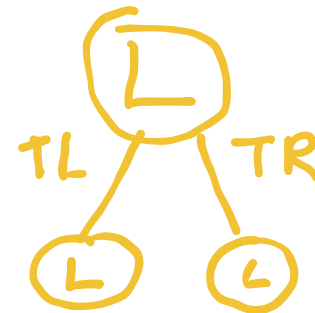
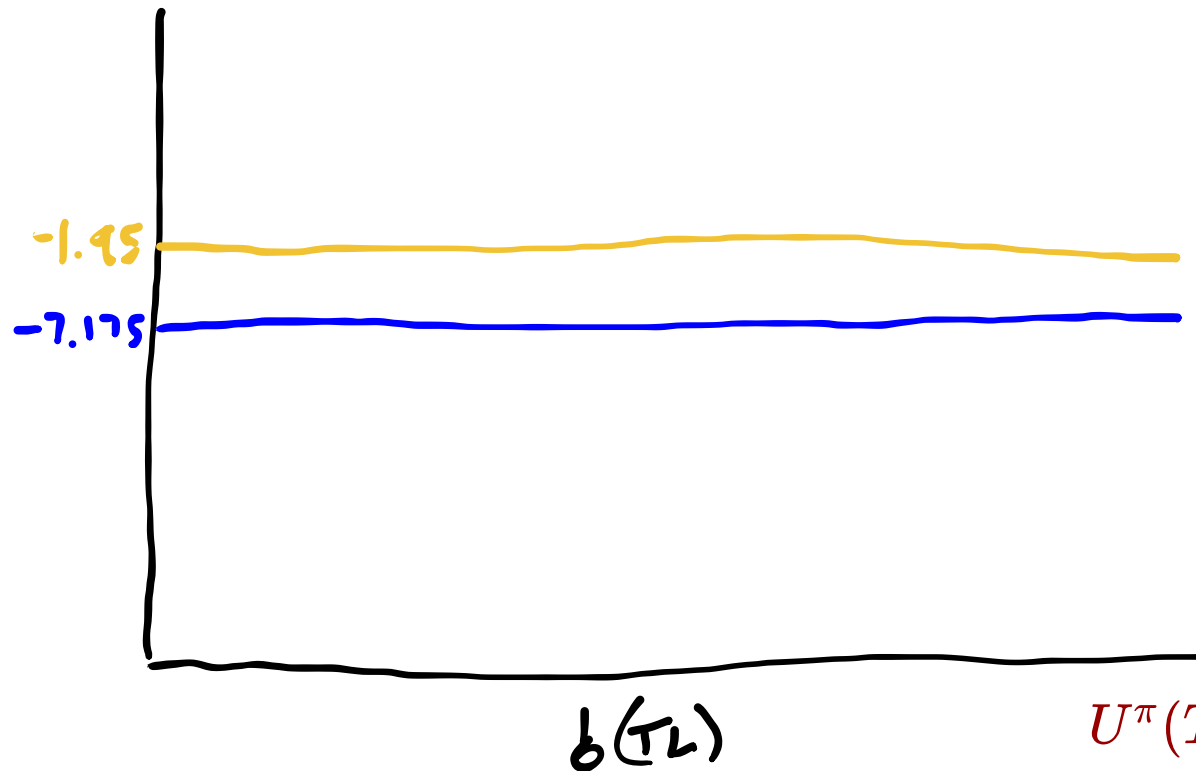
$$U^\pi(TL) = 10 + \gamma(-1)$$

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Alpha Vectors for Conditional Plans

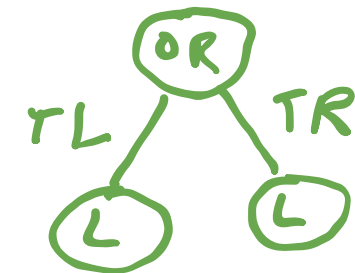
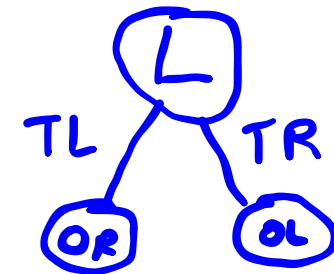
$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

For 1-step: $U^\pi(s) = R(s, \pi())$



$$U^\pi(TL) = -1 + \gamma 10$$

$$U^\pi(TR) = -1 + \gamma(-100)$$



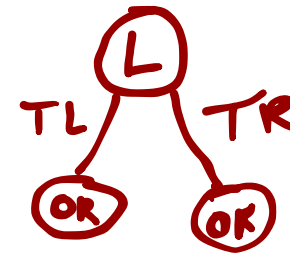
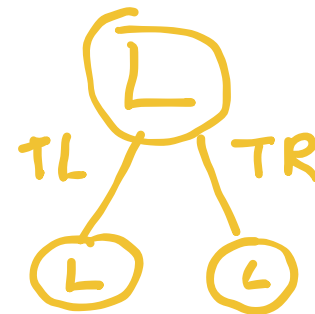
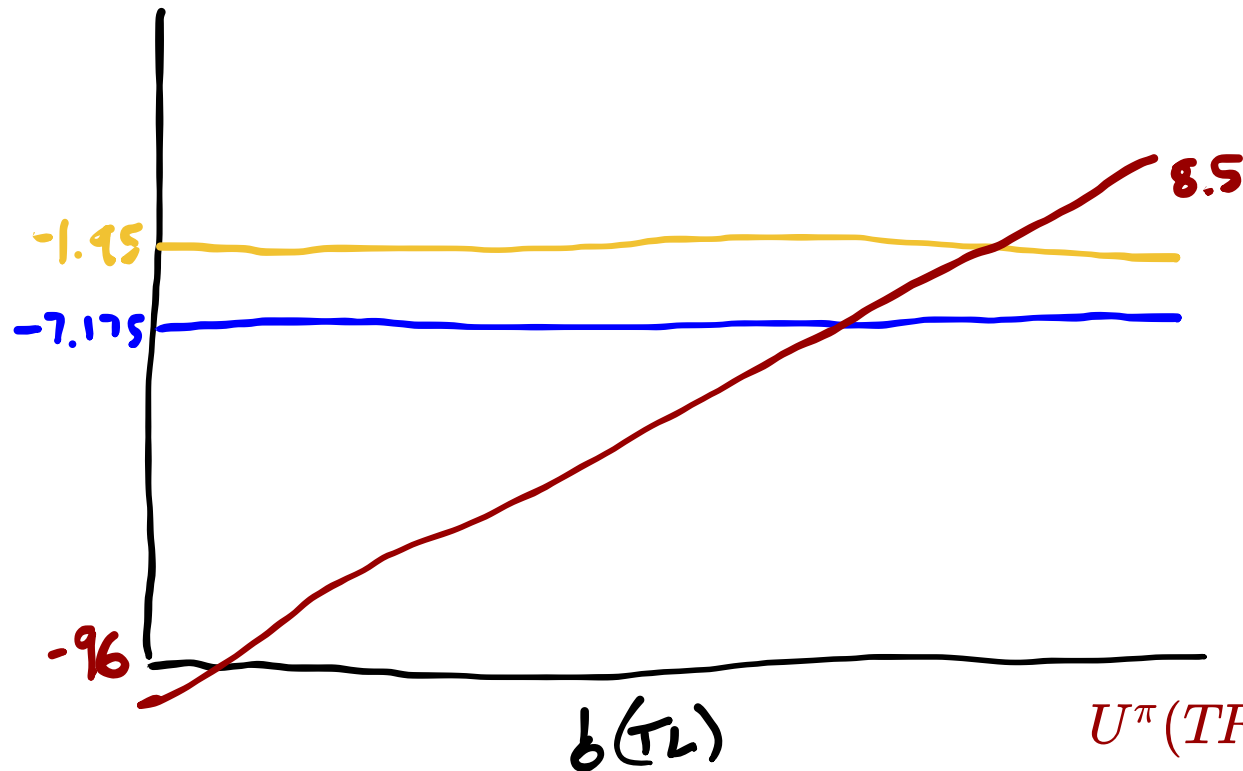
$$U^\pi(TL) = 10 + \gamma(-1)$$

$$U^\pi(TR) = -100 + \gamma(-1)$$

Alpha Vectors for Conditional Plans

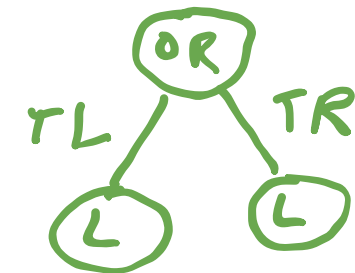
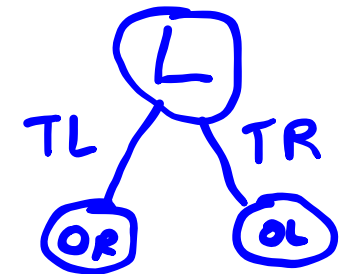
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$$U^\pi(TL) = -1 + \gamma 10$$

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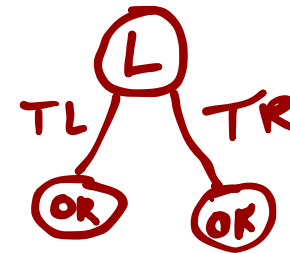
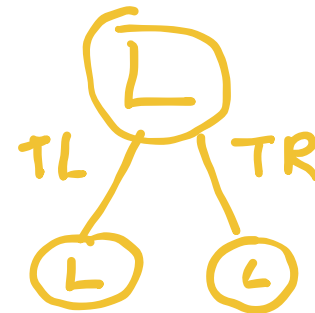
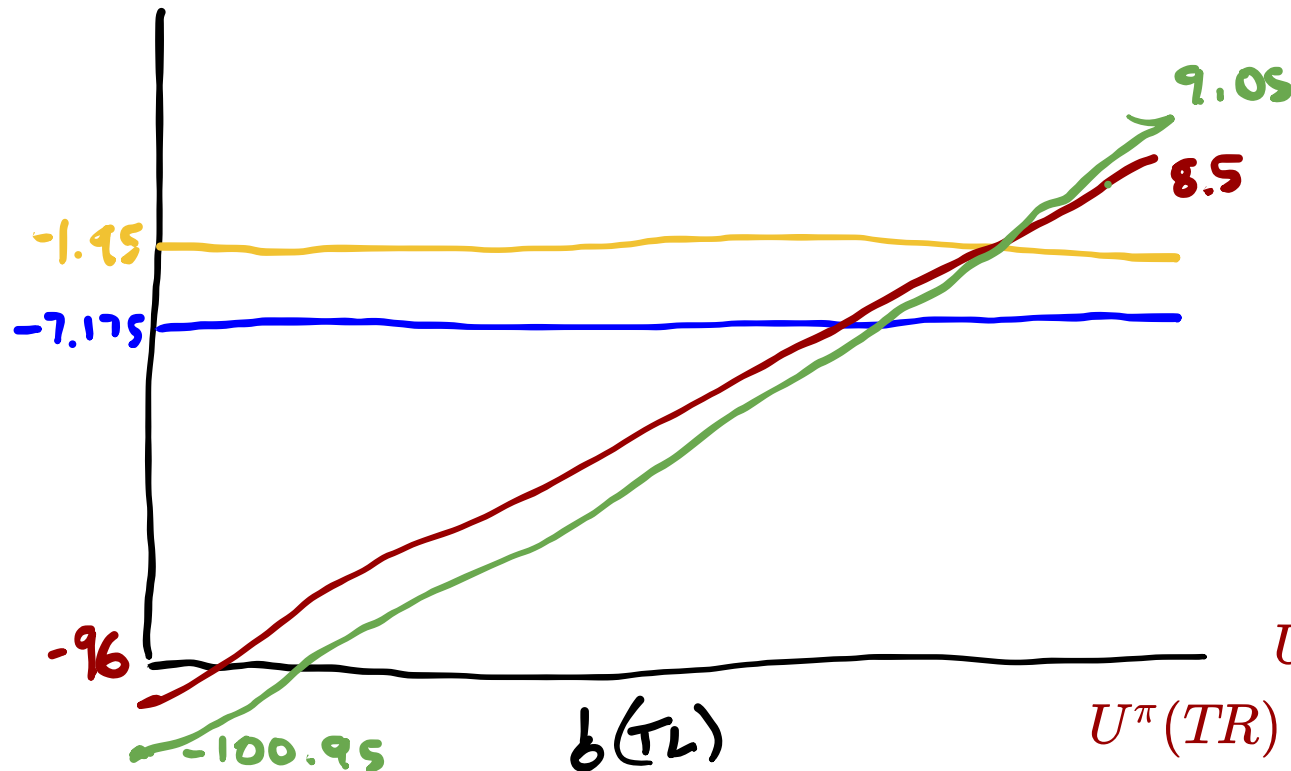
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Alpha Vectors for Conditional Plans

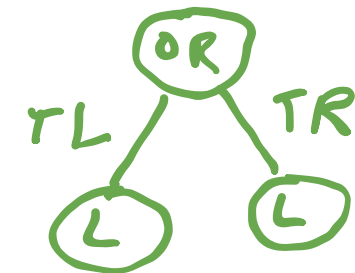
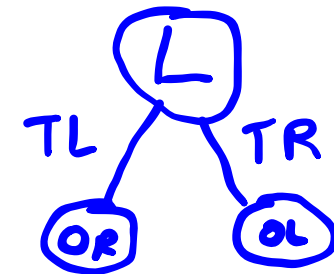
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For 1-step: $U^\pi(s) = R(s, \pi())$



$$U^\pi(TL) = -1 + \gamma 10$$

$$U^\pi(TR) = -1 + \gamma(-100)$$



$$U^\pi(TL) = 10 + \gamma(-1)$$

$$U^\pi(TR) = -100 + \gamma(-1)$$

POMDP Value Functions

POMDP Value Functions

$$V^*(b) = \max_{\alpha \in \Gamma} \alpha^\top b$$

Exercise: 2-Step Crying Baby α Vectors

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

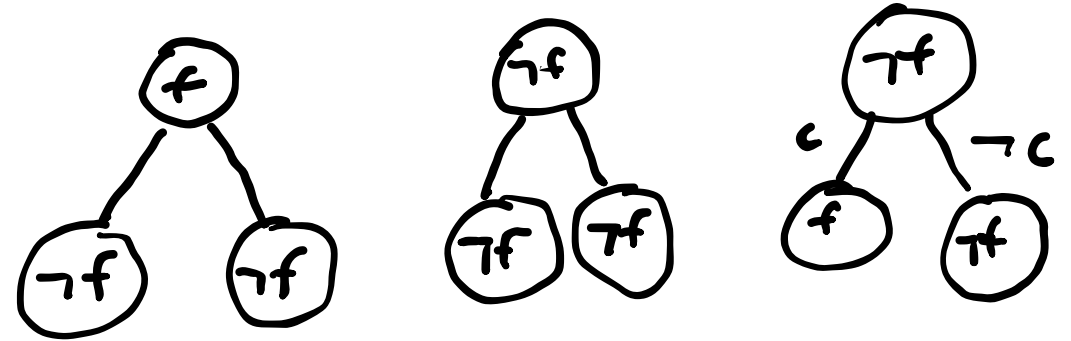
$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$



$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_o O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

α -Vector Pruning

Alpha Vector Expansion

POMDP Value Iteration (horizon d)

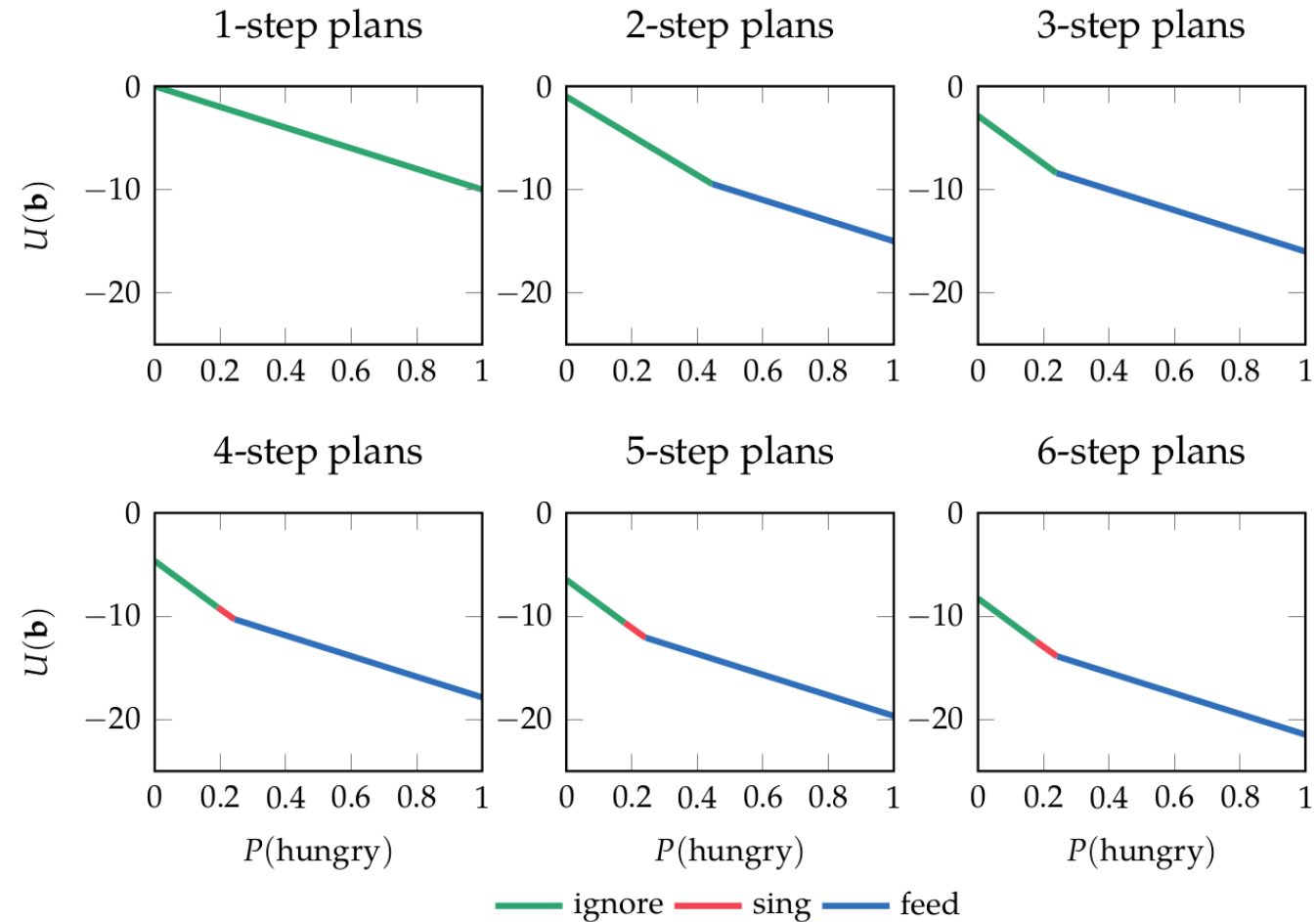
$\Gamma^0 \leftarrow \emptyset$

for $n \in 1 \dots d$

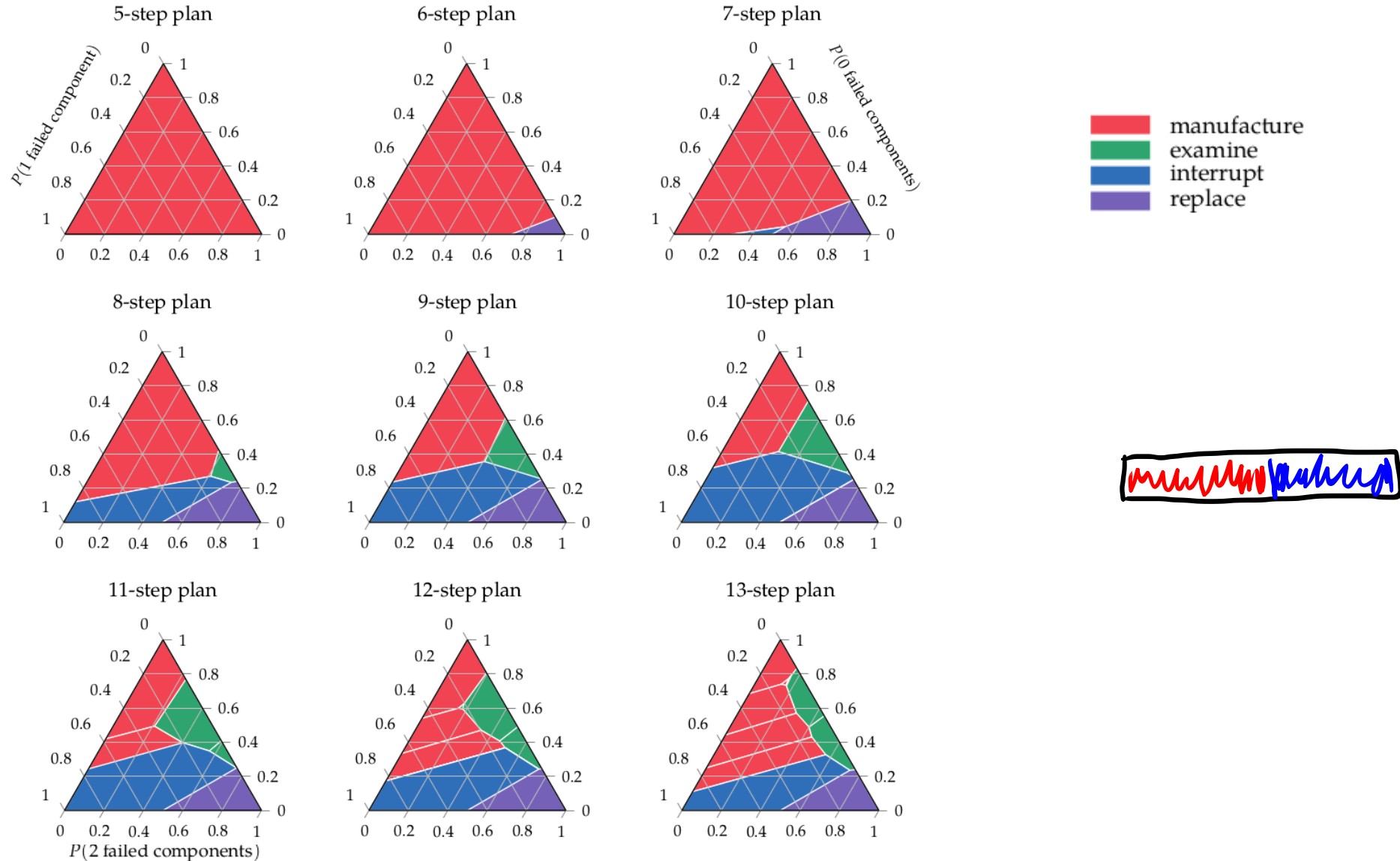
Construct Γ^n by expanding with Γ^{n-1}

Prune Γ^n

Finite Horizon POMDP Value Iteration



Finite Horizon POMDP Value Iteration



Recap

Recap

- A POMDP is an MDP on the _____

Recap

- A POMDP is an MDP on the belief space

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of _____

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors
- Each α vector corresponds to a _____

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors
- Each α vector corresponds to a conditional plan